

ZS-F42

The Frozen-Excess Hierarchy Generator and the Honest Terminus of the B3 Charge-Unit Obstruction: $\sqrt{N_4}$ Excess Theorems on the $\hat{A}_9$ Z-Spin Mediation Cycle, and the Structural (not Absolute-Scale) Closure of the Odd Susceptibility χ_-

Executing the ZS-F41 Stopping-Rule Clause (b): a Non-Transmutation Hierarchy Generator Delivered as Theorem, with the Frozen-Excess Structure Closed End-to-End and the Single Remaining Datum Isolated as the Dimensionful Membrane Charge e_6

Author: Kenny Kang **Date:** March 2026 **Theme / Paper code:** Foundations — ZS-F42 v1.9 **Series:** The Z-Spin Cosmology Papers

Verification: 101/101 PASS + 23/23 guards | Zero Free Parameters | 16 firewalled observations printed separately | companion script [zs_f42_verify_v1_9.py](#) (exits non-zero on any theorem-tier failure; no fail-open clause)

Editorial note (v1.9 = v1.8.1). This is the consolidated edition of the v1.0–v1.8 line. The abstract states only the v1.8 terminus; the per-revision history is compressed to one line each in §0.1 and detailed in Appendix C (Records 1–11). All superseded residual ledgers are collected in Appendix C.1 (Versioned Audit Trail); the body carries the single v1.9 Final Residual Ledger (§12). No result, theorem, or verification is changed from v1.8 — this is a structural/editorial compression only.

§0. Abstract

The B3 problem — the absolute vacuum-energy scale — enters Z-Spin Cosmology as the odd susceptibility χ_- , with $\rho_{\Lambda, Z} = \frac{1}{2}\chi_- \omega^2$ (F32.25–F32.27). This paper opens under the ZS-F41 Stopping-Rule clause (b) and delivers a **non-transmutation hierarchy generator as a theorem**, then drives its single residual to the corpus's honest terminus.

The theorems (unconditional). On the $\hat{A}_9$ ten-cycle carrying the strictly parity-reversing Z-Spin mediation (F41.3, PROVEN), any N_4 -step frozen mediation history decomposes into contractible even trails plus wrap–antiwrap pairs (both flux-neutral) plus $|k^*|$ **unpaired frozen windings**, and CP symmetry forces $E|k^*| = (\sigma/L)\sqrt{(2N_4/\pi)}$: the hierarchy number $\sqrt{N_4}$ **is generated as a theorem** — kinematic-statistical, not exponential (Theorems F42.2–F42.3; Monte-Carlo 1200/1200). Adjacent exponent classes are $\frac{1}{2} \ln N_4 = 277.4$ nats (≈ 120 decimal orders) apart, so the observed $v_{\text{now}} = 276.64$

selects the exponent $-1/2$ uniquely and measures the seam dimension $\mathbf{d_seam} = 1$ — the one-dimensionality of the odd seam cycle. Under the named condition (**H-FROZEN-GAS**) the projected susceptibility is the frozen-excess density $\chi_- = c_\chi \bar{M}_P H_- \partial^2$, with the **single pre-registered target** $c_\chi = 6\Omega_- \Lambda_0 / \omega^2 = 498 / (121\omega^2) = 0.8063350941$, equivalently $\chi_- Z / \alpha_{\text{patch}} = (3\Omega_- \Lambda)^2 = 4.2348$ — the closed form of the ZS-A26 O(1) bracket. Three IMPORTED-PROVEN arches (Henneaux–Teitelboim conjugacy, Bismut–Freed determinant-line holonomy, the identity $\theta_1''' / \theta_1' = -\pi^2 E_2$) supply the surrounding bridge.

The structural closure (v1.2–v1.7, DERIVED / DERIVED-CONDITIONAL). Six exploration cycles closed the *dimensionless structure* of χ_- end-to-end, each inheriting a corpus result rather than placing a new assumption: the walk is the ZS-A23 dimension-weighted mediator semigroup (v1.2); F42's coefficient **is** the ZS-A25 §5 Everpresent target as an identity (v1.4); the two residuals are the two conjugate faces of A25's $[\Lambda, \hat{T}_4] = i\hbar$ (v1.5); on the finite register the unimodular eigenvalue is the modular Hamiltonian eigenvalue $\Lambda_1 = -\ln(6/11) = \mathbf{K_Y}$ (PROVEN ZS-F19), with the von Neumann $U(n)$ extension ambiguity absent by finite-dimensionality and A19.NG2 evaded by operator-valuedness (v1.6); and the continuous-core lift (A24's F-A24.9) closes structurally via ZS-F32.9 (unique extension) + F32.26 (Friedrichs canonical extension on the compact flux circle), with Λ_1 preserved under the intertwining $E \circ \mathcal{L}_s = \mathcal{L}_s \circ E$ (v1.7). **This is operator-structure closure, not dimensionful-scale closure:** B3's *structure* is closed at both resolutions, and only its *absolute scale* remains.

The honest terminus (v1.8, the terminal NON-CLAIM). The single remaining unknown — the dimensionful $\chi_-(A, Q, \bar{M}_P) = e^{-2}/(4\pi^2 Z_-)$ — is a **genuine OPEN the corpus proves current tools cannot close**. ZS-F33's Charge-Unit Obstruction (F33.8a, CLOSED-NEGATIVE-under-R1–R3, PROVEN) establishes that flux integrality fixes the flux number k but not the dimensionful unit e^{-2}/Z_- : **(A, Q) and topology alone provably cannot determine χ_-** (Theorem F42.32). F42's tools are (A, Q)-**dimensionless** while χ_- is **dimensionful** — ZS-A25's Conditional Local-Stationary No-Go in its sharpest form (Theorem F42.33). External PROVEN math (Dirac, Gukov–Vafa–Witten, Freed–Witten) fixes dimensionless integers only and **confirms rather than lifts** the obstruction (Theorem F42.34). ZS-F34 localizes χ_- to a master quadratic form whose residual is the membrane charge $\mathbf{e}_6$, deferred to the not-yet-existent ZS-F35/F36 (Theorem F42.35). The dimensional separation is clean: in $\rho_- \Lambda_- Z = \frac{1}{2} \chi_- \omega^2$, the coefficient $\omega^2/2 = 2.5521$ is dimensionless and corpus-locked (PROVEN ZS-M1); all dimension is carried by χ_- . **The one-line verdict: F42 structurally completes the frozen-excess hierarchy but does not close B3; the remaining datum is the dimensionful membrane charge $\mathbf{e}_6$** — the sharpest, fully-localized statement of the B3-B residual, registered (not forced) exactly where the corpus's charge-unit arc (F33/F34/F35/F36/A31) places it.

The honest boundary. B3 is **not closed** (structural completion is not absolute-scale closure). The boundary odd charge unit is recovered as a **corollary, not an axiom**: $e_- \lfloor \{Z=1\} = (2\pi/\omega) \sqrt[4]{(498/121)} \bar{M}_P H_- \partial = 5.6421 \bar{M}_P H_- \partial$ — the odd charge unit is a boundary-normalized collective response that must import an IR datum. Zero fitted parameters; (A, Q, dim Z) = (35/437, 11, 2) LOCKED.

§0.1 Revision summary (one line each; details in Appendix C, Records 1–11)

- **v1.0** — Opens clause (b): the Three-Trap No-Go, the Flux-Decomposition and $\sqrt{N_4}$ Excess theorems (PROVEN), (H-FROZEN-GAS), and the single pre-registered target $c_\chi = 498/(121\omega^2)$.
- **v1.1** — Closes the two weakest legs: T-GKLS lumping (F42.12), $v_q = 1$ symmetry-locked (F42.13), the barrier band (F42.14); T3 restructured downward.
- **v1.2** — Paradigm shift: the three residuals are one already-PROVEN object, the ZS-A23 dimension-weighted mediator semigroup, inherited wholesale (F42.16–F42.18).
- **v1.3** — Correction: the seam action is π/A (ZS-M3), not $1/\kappa^2$ (ZS-A24 §14.2 retraction inherited); O-9 retracted; both residuals confirmed OPEN.
- **v1.4** — Half-closure: F42's c_χ is the A25 Everpresent target (identity, F42.19); $\alpha_{\text{patch}} = \pi_Z = 2/11$ supplies A26's missing quantity (F42.20); discrimination stays OPEN.
- **v1.5** — Conjugate-clock unification: the two residuals are the two faces of A25's $[\Lambda, \hat{T}_4] = i\hbar$ (F42.21); F42 supplies the Λ -sector operator A26 lacked; residual $2 \rightarrow 1$.
- **v1.6** — Finite-register closure: $\Lambda_1 = -\ln(6/11) = K_Y$ (PROVEN ZS-F19); $U(n)$ ambiguity absent by finite-dimensionality; A19.NG2 evaded (F42.24–F42.27).
- **v1.7** — Continuous-core closure (structural): F32.9 unique extension + F32.26 Friedrichs canonical extension; Λ_1 preserved under E-intertwining (F42.28–F42.31).
- **v1.8** — The honest terminus: the dimensionful χ_- is a genuine OPEN the corpus proves current tools cannot close; terminal NON-CLAIM, residual = membrane charge e_6 (F42.32–F42.35).
- **v1.9 (= v1.8.1)** — Editorial consolidation: abstract compressed to the v1.8 terminus, per-revision history to one line each, all superseded ledgers moved to Appendix C.1. No result changed.

Epistemic Status Legend

Tag	Meaning
PROVEN	Mathematical theorem with complete proof inside this paper or machine-verified identity.
IMPORTED-PROVEN	External published theorem consumed as-is (citation given); its <i>placement</i> in Z-Spin may carry a separate conditional tag.
DERIVED	Follows from locked corpus inputs by explicit computation, no new conditions.
DERIVED-CONDITIONAL	Derived contingent on named, registered conditions (listed in the Condition Registry, §3.4).
HYPOTHESIS-strong	Convergent under deep-exploration cycles, corpus-consistent, anti-numerology MC pre-registered; not yet a theorem.

Tag	Meaning
COMPUTED / COMPUTED-INCOMPLETE	Numerical result under a stated normalization; incomplete if the normalization is not corpus-fixed.
OBSERVATION (FIREWALL)	Numerical proximity recorded for the ledger; never consumed as evidence (inherits F-F32.27, F-F40.8, F-F41.7).
OPEN	Genuinely undetermined within current corpus tools.
NON-CLAIM	Explicitly not asserted.
RETIRED	A prior formulation replaced by a sharper one (recorded, not erased).

§1. Introduction: lineage, the terminus, and what this paper is

Lineage. Consumes ZS-F41 v1.1, ZS-F40 v1.2, ZS-F36 v2.1, ZS-F35 v1.5, ZS-F34 v1.8, ZS-F33 v1.8, ZS-F32 v1.5, ZS-F31 v1.4, ZS-A32 v1.1, ZS-A30 v2.1, ZS-A28, ZS-A26, ZS-A25 v1.6, ZS-M46 v1.4, ZS-M8, ZS-M1. External imports: Henneaux–Teitelboim [1], Sorkin and the Everpresent- Λ programme [2–5], Leutwyler–Smilga [6], Quillen [7], Bismut–Freed [8], Bismut–Gillet–Soulé [9], Aizenman–Lebowitz–Ruelle [10], Dasgupta–Ma [11], Fisher [12], Kaloper–Padilla [13], Kaloper–Sorbo [14], Pólya [15], Khinchin [16], Witten [17], Bousso–Polchinski [18], Planck 2018 [19].

Where the corpus stands. After ZS-F40 v1.2 the B3 frontier is a single object: the odd susceptibility χ_- , with $\rho_{\Lambda,Z} = \frac{1}{2}\chi_- \omega^2$ (F32.25–F32.27 under C1–C2), its category fixed by the Zero-to-Susceptibility Boundary Principle (Thm F40.8: $\theta_1(0|\tau_K) = 0$, $\theta_1'(0|\tau_K) = 2\eta^3 = 14.14653 \neq 0$, the Quillen determinant-line section that survives the zero), every corpus route to its absolute value CLOSED-NEGATIVE or tautological (F40 §7), and the Charge-Unit Obstruction PROVEN-irreducible (F33 §9.4–9.5: flux integrality fixes the flux number, never the unit). ZS-F41 supplied the first conditional odd charge lattice ($A_9 \rightarrow C_5$ under (H-SPIN)), proved the Bipartite Mediation-Parity Theorem (F41.3: the McKay graph of Z_{10} is the ten-cycle $\hat{A}_9$; Z-Spin mediation $\otimes 2$ is strictly parity-reversing; the color classes are the seam grading), closed both transmutation stages, and registered the Terminal Stopping Rule with the Quillen-curvature susceptibility as the *named, not begun* clause-(b) candidate (NC-F41.8).

What this paper is. Three consecutive deep-exploration cycles (records in Appendix C) converged on a single derivation skeleton — the **QCS–HT identification**: the Quillen-curvature susceptibility of F41 §11(b) and the Everpresent four-volume fluctuation of A25 §5 Escape 2 are the *same object* seen in two languages, tied by the Henneaux–Teitelboim conjugacy $[\Lambda, \hat{T}_4] = i\hbar$ that the corpus already carries in F32.24's compact odd three-form. The third cycle attempted the proof of the keystone and returned a split verdict which this paper freezes: the connecting *slogan* fails as stated (three traps, §5), but the connecting *mechanism* — the frozen-excess dilute gas — yields two unconditional theorems (§6–§7) and one sharply conditional identity (§8). The paper therefore delivers a **non-transmutation hierarchy generator as a**

theorem (the $\sqrt{N_4}$ excess law), the item clause (b) names, while stating plainly that the *transfer* of that generator into the value of χ_- remains conditional.

What this paper is not. It is not a closure of B3 (NC-F42.1). It fixes no dimensionful scale from **A**, **Q**, and topology alone — the IR boundary datum H_{∂} enters explicitly, as F-F40.8 (Buckingham- π) requires, and the paper's central structural claim is precisely that this import is *forced*: the odd charge unit is a boundary-normalized collective response (§11.4).

§2. Stopping-Rule compliance audit (F-F41.10)

ZS-F41 §11 (binding): *"No successor B3 paper is opened unless the corpus first derives, as a theorem from the axioms, either (a) the odd coupling boundary value $\alpha_- \neq \kappa^2$, or (b) a non-transmutation hierarchy generator — the named candidate being the Quillen-curvature susceptibility $\chi_- = -\partial^2 J \log Z|_0$ on the A_9/C_5 determinant line... Theorem first; paper second; otherwise silence."* ZS-F41's change log additionally retired "the proposed successor paper (F42)." This audit is therefore mandatory and is placed before any result.

2.1 What clause (b) demands. A hierarchy generator that (i) is not an exponential transmutation $e^{(-S)}$ (both frozen registrations of that family are CLOSED-NEGATIVE, F41 §7–§8), (ii) arrives as a theorem from the axioms, and (iii) lands on the named object — the Quillen-curvature susceptibility.

2.2 What is delivered as theorem. Theorems F42.2 and F42.3 (§6–§7) prove, from (α) the PROVEN bipartite parity-reversing structure of Z-Spin mediation on $\hat{A}_9$ (F41.3), (β) CP symmetry of the register increments (mean-zero), and (γ) classical probability (CLT, Pólya, Khinchin — IMPORTED-PROVEN), that a frozen N_4 -step mediation history carries an unpaired winding excess of size $|k^*| = O(\sqrt{N_4})$, with $E|k^*| = (\sigma/L)\sqrt{(2N_4/\pi)}$. The generator is $\sqrt{N_4}$ — the size of the causal register, "small because the universe is old" (A25 §5) — a **kinematic-statistical** generator, category-disjoint from transmutation. Criterion (i) ✓ and (ii) ✓ for the *generator*. Criterion (iii): the generator is located exactly on the F41 A_9 lattice and, via the Bismut–Freed holonomy theorem [8] (the determinant-line holonomy as the adiabatic limit of the exponentiated η -invariant), the accumulated frozen winding *is* a holonomy datum of the determinant line whose curvature is the Quillen second variation [7, 9] — the named object. Placement of (iii) is DERIVED-CONDITIONAL on (H-CYC) (§3.4).

2.3 What is not delivered as theorem. The identity " χ_- equals the response of the frozen excess" — the value transfer — is Proposition F42.5, DERIVED-CONDITIONAL on (**H-FROZEN-GAS**) (§8). The value of χ_- therefore remains conditional.

2.4 Dual-reading verdict. Under the **generator-reading** of clause (b) ("derive the hierarchy generator as a theorem"), the gate opens: this paper is the clause-(b) execution, delivering the theorem pair plus the conditional identification. Under the **strict value-reading** ("derive the χ_- value as a theorem"), the gate remains closed: this paper is then the registered theorem ledger and pre-registration document, its §8–§9 identification frozen at the conditional tier, and B3's status is unchanged. Both readings are recorded; the

collaboration's adoption of the strict reading fires gate **F-F42.0** (§14), which demotes §8–§9 to HYPOTHESIS-strong-tier claims while §§5–7 theorems stand unconditionally. **Neither reading closes B3** (NC-F42.1).

§3. Setup: locked inputs, imports, the walk model, and the Condition Registry

3.1 Locked inputs (zero free parameters). $A = 35/437$, $Q = 11$, register $(Z, X, Y) = (2, 3, 6)$, $\kappa^2 = A/Q = 35/4807$. The i -tetration fixed point $z^* = 0.4382829367 + 0.3605924719i$ ($f(z) = i^z z$; Banach, $|\lambda^*| = 0.8915 < 1$), multiplier $\lambda^* = (i\pi/2)z^*$, $\omega = \arg \lambda^* = 2.2592495540$, $\mu = -\ln|\lambda^*| = 0.1148346250$, Koenigs modulus $\tau_K = (\omega + i\mu)/2\pi$ with the Nome Identity $q(\tau_K) = \lambda^*$ (F40.L2, PROVEN; check A5). Vacuum fraction $\Omega_{\Lambda,0} = 83/121$ (ZS-A30, DERIVED-CONDITIONAL — consumed as a conditional input, untouched). $\omega^2/2 = 2.5521042734$.

3.2 Corpus imports (verbatim anchors). (i) $\rho_{\Lambda,Z} = \frac{1}{2}\chi\omega^2$ under **C1** (canonical quadratic / large- N branch law [17]) and **C2** ($\theta_Z = \omega$), with $k^* = 0$ selected by $0 < \omega < \pi$ — F32.25–F32.26. (ii) The compact odd $U(1)$ three-form: Dirac-quantized flux $k \in \mathbb{Z}$, monodromy protection [14] — F32.24 (DERIVED-CONDITIONAL). (iii) Selective sequestering of the even sector: "even vacuum-energy loops, including virtual gravitons, are removed from the curvature source by the Kaloper–Padilla *Omnia Sequestra* mechanism" [13] — F32.13 (IMPORTED-PROVEN + DC). (iv) Central-Shift Orthogonality: relative/modular tools are blind to the absolute offset — F32.15 (PROVEN). (v) Zero-to-Susceptibility: $\chi = -\partial^2_J \log Z(J)|_0$ after zero-mode projection; $\theta_1'(0|\tau_K) = 2\eta^3 = 14.14653$, the Quillen section surviving the zero — F40.8. (vi) Bipartite Mediation-Parity: $\text{McKay}(Z_{10}) = \hat{A}_9$ (ten-cycle); Z -Spin mediation $\otimes 2$ strictly parity-reversing; bipartite color classes = seam grading; "a McKay cycle admits the seam grading iff it is even iff the binary lift is present" — F41.3 (PROVEN), under (H-SPIN). (vii) The normalizable causal-entropic present measure $d\mu_{ZCD} \propto A_D(T)\sigma_Z(T)dT$ with the entropy-budget bound; spectral gap $\Delta = 2A/Q$ — F31 §10.2 (DERIVED). (viii) The Everpresent bypass: $\Delta\Lambda \sim V_4^{(-1/2)}$, "there is no $e^{(-276)}$ to compute — small because the universe is old"; $O(1)$ target $\Omega_{\Lambda,\text{rms}} = (1/3)\sqrt{(\chi_Z/\alpha_{\text{patch}})}$; three named risks (sign, $w(z)$, universality); χ_Z bracket $(0.2, 2.1)$ on the $(3, 2, 6)/11$ bottleneck generator — A25 §5, A26 (OPEN / COMPUTED-INCOMPLETE). (ix) $\rho_{\Lambda}^{(1/4)} = \sqrt{(\tilde{M}_P H_0)}$: the geometric mean as "a localization of the naturalness problem, not a resolution"; the present-epoch identification conditional U_N — A28 Part II.

3.3 The walk model (placement). The frozen odd register history is modeled as an N -step nearest-neighbour walk on the cycle graph C_L , $L = 10$ (the $\hat{A}_9$ node cycle), with i.i.d. CP-symmetric increments $\varepsilon_n \in \{\pm 1\}$ ($\langle \varepsilon \rangle = 0$, variance σ^2), lifted to $W_n = \sum \varepsilon_i \in \mathbb{Z}$ on the universal cover. The compact odd flux integer of F32.24 is identified with the winding $k = W/L$ (up to the open-trail remainder $|d| < L$). The θ_Z -angle couples to k as $e^{(ik\theta)}$ (Leutwyler–Smilga sector-Fourier structure [6], the same skeleton as the F32.26 spectrum $(2\pi k + \omega)^2$). This model is a *placement* of the PROVEN F41.3 structure, not a derivation from the 121-face GKLS generator — obligation **T-GKLS** (§12). Theorems F42.2–F42.3 are unconditional mathematics about such walks; their physical force inherits the registry below.

3.4 Condition Registry.

Condition	Content	Origin	Status
(H-SPIN)	seam parity $\beta = \text{central } \{\pm 1\}$ of Z-Spin SU(2); grounds $\hat{A}_9/C_5$	ZS-F41 (inherited)	HYPOTHESIS-strong
C1, C2	quadratic/large-N law; $\theta_Z = \omega$	ZS-F32.25 (inherited)	DERIVED-CONDITION AL
(H-GATE) $\equiv U_N$	V_4 frozen at the F31 causal-entropic present gate; identical to the A28-D present-epoch conditional	ZS-F31/A28 (inherited; no new condition)	DERIVED-CONDITION AL
(H-CYC)	the F32.24 flux integer = the winding of the Z-Spin mediation walk on $\hat{A}_9$ = the holonomy count of the Bismut–Freed determinant-line transport along the odd seam one-cycle of E^*	this paper (new placement)	HYPOTHESIS-strong
(H-FROZEN-GAS)	three clauses, §8; in v1.1 clause (b) $\rightarrow$ (b') independence only ($v_q = 1$ PROVEN), clause (c) $\rightarrow$ band DERIVED , exact S_{wrap} OPEN	this paper (new mechanism)	(a) anchored F32.13; (b') HYPOTHESIS-strong; (c) DERIVED-CONDITION AL
(C-DAV)	secular / weak-coupling validity for F42.12	v1.1	v1.2: CLOSED — Thm F42.16 inherits the PROVEN A23.14a generator; no secular limit
(C-ATT)	attempt frequency fixing exact S_{wrap} in-band	v1.1	v1.2: reduced — = A23 microscopic scale Γ_0 (named); exact in-band value OPEN

§4. The object and the strategy

The chain to be assembled is $\mathbf{p}_{\Lambda, Z} = \frac{1}{2}\chi\omega^2$ [F32] $\rightarrow \chi_-$ = **second response after zero-mode projection** [F40.8] $\rightarrow \chi_-$ = **frozen-excess density** [this paper] $\rightarrow \chi_- = \mathbf{c}_{\chi} \tilde{\mathbf{M}}_P^2 \mathbf{H}_\partial^2$, $\mathbf{c}_{\chi} = 6\Omega_\Lambda/\omega^2$ [conditional identification]. The mathematical crux is the third arrow: *which* ensemble's second response, evaluated *where*, produces the exponent $-1/2$ in N_4 . Three natural readings fail (§5); the fourth (§6–§8) is the paper's content. Throughout, the four arches of the surrounding bridge are IMPORTED-PROVEN: the HT conjugacy $[\Lambda, \hat{T}_4] = i\hbar$ [1] (carried in the corpus by F32.24), the fluctuation–response/sector-Fourier

skeleton [6], the Quillen/Bismut–Freed determinant-line geometry [7–9], and the classical E_2 identity of §10.

§5. Proposition F42.1 — the Three-Trap No-Go [PROVEN within the model]

Proposition F42.1. In the walk model of §3.3, none of the following operations produces $\chi \propto N^{-(1/2)}$:

(A) Annealed: summing freely over histories, $-\partial^2_{\theta} \log Z_{\text{ann}}|_0 = \text{Var}(k) = \sigma^2 N/L^2$ — *extensive*.

(Verified: $\text{Var}(k) = 9.9, 102, 1012, 10076, 100846$ at $N = 10^3 \dots 10^7$; exactly $N/100$; checks E4–E5.) This is the cosmological-constant problem restated, and — diagnostically — the ensemble whose constant every one of the five F40 §7 routes was implicitly targeting (§11.3). **(B) Fully frozen:** a completely frozen history contributes the pure phase $e^{(ik \cdot \theta)}$; $-\partial^2_{\theta} \log Z^* \equiv 0$ — the susceptibility vanishes identically (check F1). **(C) Bridge-conditioned:** conditioning on closure ($W \equiv 0 \pmod{L}$) leaves $\text{Var}(k \mid \text{bridge}) \approx \sigma^2 N/L^2$ — still extensive (check F2: 101.7 vs 100 predicted).

Proof. (A) is the binomial variance; (B) is the second derivative of a linear function; (C) follows since conditioning on a residue class mod L changes the variance by $O(1)$.

Consequence [RETIRED]. The slogan of the preceding exploration cycle, "zero-mode projection = quenching," is hereby retired as a candidate theorem: it does not, as stated, select any operation lying between traps (A) and (B). The $\sqrt{N}$ law requires the *number of responding degrees of freedom* to be $\sqrt{N}$ — neither N (trap A) nor 0 (trap B). Identifying those degrees of freedom is Theorem F42.2.

§6. Theorem F42.2 — the Frozen-History Flux Decomposition [PROVEN]

Theorem F42.2. Let γ be an N -step nearest-neighbour walk on the cycle C_L with L even ($\hat{A}_9$: $L = 10$; bipartite by F41.3), lifted to $W \in \mathbb{Z}$. Then: **(i) Parity.** Every closed sub-walk (same start and end node) has even length, hence trivial seam $\mathbb{Z}_2$ charge. (Return $\Leftrightarrow \Delta W \equiv 0 \pmod{L}$; L even $\Rightarrow \Delta W$ even; $\Delta W \equiv \text{length} \pmod{2}$.) **(ii) Decomposition and homology additivity.** Cutting γ at the successive visit times $0 = t_0 < t_1 < \dots < t_r$ of (lifts of) the start node yields $\gamma = \gamma_1 \circ \dots \circ \gamma_r \circ \gamma_{\text{open}}$, where each closed γ_j carries the integer winding $w_j = (W_{\{t_j\}} - W_{\{t_{j-1}\}})/L \in \mathbb{Z}$ and $L \cdot \sum_j w_j + d_{\text{open}} = W_N$, $d_{\text{open}} = W_N - W_{\{t_r\}}$. **(iii) Winding cost.** Any closed trail of winding w has length $\geq L|w|$; hence the steps carried by winding trails number $\geq L \cdot \sum_j |w_j|$. **(iv) Pair structure of the flux content.** The multiset $\{w_j\}$ decomposes into $(+1, -1)$ **wrap–antiwrap pairs** — flux-neutral composites — plus $|k_N|$ **unpaired windings of a single sign**, where $k_N = \sum_j w_j + O(1)$ is the net flux. The θ_Z -coupled charge of the entire frozen history is carried exclusively by the unpaired windings; the contractible trails ($w = 0$) and the wrap–antiwrap pairs are θ_Z -neutral **identically** ($e^{(i \cdot 0 \cdot \theta)} = 1$ for all θ).

Proof. (i) Bipartiteness of even cycles; each mediation step flips the seam color (F41.3(ii), PROVEN). (ii) Telescoping of the lift coordinate; each segment between returns has $\Delta W \in L\mathbb{Z}$ by definition of return on

C_L. (iii) The lift changes by at most one per step. (iv) Minimal sign-pairing of an integer multiset; a paired $(+1, -1)$ composite has total winding zero, hence couples to θ as $e^{i0\theta}$.

Machine verification (Appendix B; checks D1–D6). 200 random histories at each of $N = 1000$ and $N = 5000$, seed 11: even closed-trail lengths $200/200 + 200/200$; homology additivity $200/200 + 200/200$; winding-cost bound $200/200 + 200/200$ — **1200/1200**.

Remark (the two charges — a correction frozen here). The seam charge $\beta \in \mathbb{Z}_2$ and the flux $k \in \mathbb{Z}$ are distinct: a full wrap is β -even but k -charged; a contractible even trail is both β -even *and* k -neutral. θ_Z couples to k . The statement "seam-even $\Rightarrow$ θ -inert" from an earlier exploration cycle is corrected to "**k-neutral $\Rightarrow$ θ -inert identically**; β -even, k -neutral composites are *additionally* F32.13-sequestered from the curvature source." The correction is recorded in Appendix C, Record 3.

§7. Theorem F42.3 — Excess Statistics [PROVEN / IMPORTED-PROVEN, verified]

Theorem F42.3. For CP-symmetric i.i.d. increments ($\langle \varepsilon \rangle = 0$, $\text{Var } \varepsilon = \sigma^2$): **(i)** $W_N/(\sigma\sqrt{N}) \Rightarrow \mathcal{N}(0, 1)$ and $E|W_N| = \sigma\sqrt{(2N/\pi)(1 + o(1))}$ — the half-normal mean. **(ii)** The unpaired-winding count satisfies $E|k_N| = (\sigma/L)\sqrt{(2N/\pi)(1 + o(1))}$. **(iii)** (Khinchin LIL [16]) $\limsup |W_N|/\sqrt{(2\sigma^2 N \ln \ln N)} = 1$ almost surely — the a.s. upper envelope. **(iv)** (Pólya recurrence [15]) $\liminf |W_N| = 0$ almost surely.

Proof. Classical: (i)–(ii) CLT and the half-normal first moment; (iii) Khinchin; (iv) one-dimensional recurrence. **Verification (checks E1–E3):** $E|W|/\sqrt{N} = 0.8012$ ($N = 10^4$), 0.8046 ($N = 10^6$) against $\sqrt{(2/\pi)} = 0.7979$; $E|k^*| = 2.51, 8.05, 25.4, 80.1, 253.2$ at $N = 10^3 \dots 10^7$ against $\sqrt{(2N/\pi)}/10 = 2.52, 7.98, 25.2, 79.8, 252.3$.

Corollary F42.3a (sign-crossings and the honest absorption of "why now"). By (iv) the frozen net flux — hence the frozen ρ_Λ — passes through zero at some epochs: the Z-Spin realization *inherits the fluctuating character* of the Everpresent- Λ phenomenology [2–4] rather than contradicting it. The probability that the present realization satisfies $|\xi^*| \geq 0.8$ (the half-normal variable $\xi = |W|/\sigma\sqrt{N}$ at the value implied by $\Omega_\Lambda, 0 = 83/121$, modulo the $O(1)$ register factors of §8) is $P = \text{erfc}(0.8/\sqrt{2}) = 0.424$ — an **O(1) event requiring no tuning** (firewalled observation O-4). The *specific* value of ξ^* is one number, and it is filed where the corpus already files it: the U_N present-epoch conditional (A28-D). **No new coincidence condition is created.**

§8. (H-FROZEN-GAS) and Proposition F42.5 — the conditional susceptibility identity

Definition F42.4 — the named condition (H-FROZEN-GAS). **(a) Inert bulk.** The contractible trails and wrap–antiwrap pairs of the frozen history are θ_Z -inert (this half is Theorem F42.2(iv), *identical* vanishing) and, as seam-even k -neutral composites, are removed from the curvature source by the F32.13

sequestering [13]. [*Half PROVEN here, half inherited DC.*] **(b) Live excess.** The $|k^*|$ unpaired frozen windings are the θ_Z -active quanta; each retains a unit-strength (± 1 , compact- $U(1)$ -integral, F32.24) residual fluctuation, independently in the dilute regime, so that $\text{Var_res}(k) = v_q \cdot |k^*|$ with $v_q = O(1)$. [*The dilute-topological-gas identity — susceptibility = density of active integer charges (Poisson) — is the model-class anchor; the random-singlet genre [11, 12] is the quenched-disorder precedent: pairs lock into inert singlets and the low-temperature susceptibility is carried by the unpaired free spins.*] **(c) No-nucleation.** Creation of new wraps by the gate-epoch dynamics is suppressed by $\geq \frac{1}{2} \ln N_4 = 277.4$ nats. [*Consistency gate against the F31 spectral gap $\Delta = 2A/Q = 0.014562$: required modular duration $\tau \gtrsim 1.9 \times 10^4$ units; registered, not evaluated — firewalled observations O-2, O-3.*]

Proposition F42.5 (Frozen-Gas Susceptibility Identity) [DERIVED-CONDITIONAL on (H-FROZEN-GAS)]. Under Definition F42.4, the second response of the residual (post-projection) ensemble around the frozen background is $\chi_- = \text{Var_res}(k)/V_4 = v_q |k^*|/V_4$, and with Theorem F42.3(ii) and N steps $= \alpha_{\text{patch}} N_4$, $\chi_- = c \cdot \bar{M}_P^4 / \sqrt{N_4}$, $c = (v_q \sigma/L) \sqrt{(2/\pi)} \cdot \xi^* \cdot \sqrt{\alpha_{\text{patch}}} \cdot \gamma$ -factors — all $O(1)$ register constants, to be computed blindly at T3. The three traps are avoided by construction: only $|k^*| \sim \sqrt{N}$ degrees of freedom respond (not N : trap A); the wraps retain ± 1 residual freedom, so $\text{Var_res} \neq 0$ (trap B); no closure conditioning is imposed (trap C). The sign is automatic: $\text{Var_res} \geq 0$.

Reinterpretation of F40.8 [DERIVED-interpretation]. The zero mode is the global θ -rotation; $\det'$ removes it, i.e. fixes the θ -frame in which the frozen *linear* term (the pure phase $e^{ik^*\theta}$ of trap B) is separated from the residual *quadratic* response. F40.8's category statement — "the survivor is a susceptibility" — is realized concretely: **the survivor is $\text{Var_res}(k)/V_4$** . The Leutwyler–Smilga fluctuation–response identity is retained *within* the frozen background.

§9. Theorem F42.6 and Proposition F42.7 — the assembled T1 and the QCS–HT identification

Theorem F42.6 (the $\sqrt{N_4}$ Hierarchy Transfer; DERIVED-CONDITIONAL on (H-FROZEN-GAS), (H-CYC)). Combining the HT conjugacy arch [1] (F32.24 placement), the determinant-line arch [7–9] (the frozen winding as Bismut–Freed holonomy data; χ_- as the Quillen second variation at the frozen point), the dimensionless arch (§10), and Theorems F42.2–F42.3 with Proposition F42.5: $\chi_- = c \cdot \bar{M}_P^4 / \sqrt{N_4} = c \cdot \bar{M}_P^2 H_- \hat{\partial}^2$ ($N_4 = \alpha_{\text{patch}}^{-1}$ -normalized register four-volume, frozen at the F31 gate). This is the T1 statement of the exploration cycles, promoted from HYPOTHESIS-strong to DERIVED-CONDITIONAL: the unproven core has been localized from a slogan to the two clauses (H-FROZEN-GAS)(b), (c).

Proposition F42.7 (Everpresent Identification / the QCS–HT chain; DERIVED-CONDITIONAL on the full Registry of §3.4). With $\rho_{\Lambda,Z} = \frac{1}{2} \chi_- \omega^2$ (C1–C2) and $\Omega_{\Lambda,0} = 83/121$ (A30):

$\rho_{\Lambda} = \frac{1}{2} c_{\chi} \omega^2 \bar{M}_P^2 H_- \hat{\partial}^2 = 3 \Omega_{\Lambda} \bar{M}_P^2 H_- \hat{\partial}^2$, with the **single pre-registered coefficient target**

$c_{\chi} = 6 \Omega_{\Lambda,0} / \omega^2 = 498 / (121 \omega^2) = 0.8063350941$ (checks B1–B2),

equivalently, in the A25 §5 normalization, $\chi_Z/\alpha_{\text{patch}} = (3\Omega_\Lambda)^2 = (249/121)^2 = 4.2347517246$ (check B3) — the exact closed form of the target A25 recorded as " ≈ 4.23 ", consistent in magnitude with the A26 bottleneck bracket $\chi_Z \in (0.2, 2.1)$ (COMPUTED-INCOMPLETE; the complete comparison is T3).

The three A25 risks, resolved within the chain: (a) Sign. $\rho_\Lambda = \frac{1}{2}\chi\omega^2$ is (non-negative density) $\times (\omega^2 > 0)$: positive-definite structurally. The ensemble mean $\langle \Lambda \rangle = 0$ of the Everpresent programme is a statement about *realizations*; the observable here is the response of *one frozen realization*, entering through $\text{Var}_{\text{res}} \geq 0$ and the $k^* = 0$ circle-distance minimization of F32.25. [DERIVED-CONDITIONAL, C1–C2 inherited.] **(b) w(z).** V_4 is frozen at the normalizable F31 causal-entropic present gate $\Rightarrow \rho_\Lambda$ constant $\Rightarrow \mathbf{w} = -\mathbf{1}$ exactly, consistent with the A28 import. The cost is exactly (H-GATE) $\equiv U_N$ — a condition the corpus **already carries**; residual count does not grow. [DERIVED-CONDITIONAL on U_N .] **(c) Universality.** The Z-Spin-specific content is the closed-form target above, with $\omega = \arg \lambda^*$ and $\Omega_\Lambda = 83/121$ both locked — a sharp, falsifiable number replacing a generic scaling claim. [Pre-registered; F-F42.8.]

§10. The dimensionless arch: $\theta_1'''/\theta_1' = -\pi^2 E_2$ and the Koenigs fiber [IMPORTED-PROVEN, verified]

The projected second variation on the internal Koenigs fiber is governed by the classical identity (πz convention) $\theta_1'''(0|\tau)/\theta_1'(0|\tau) = -\pi^2 E_2(\tau)$, verified numerically in the companion script in the unscaled-z convention (ratio = $-E_2$): at $\tau = i$, $E_2(i) = 3/\pi$ (check G1–G2); at τ_K , direct q-sum and S-transform agree to 10^{-8} , $E_2(\tau_K) = 71.8127 - 52.9200i$, ratio + $E_2 < 10^{-6}$ (checks G3–G4); $|\theta_1'(0|\tau_K)| = 14.14653 = |2\eta(\tau_K)^3|$ (checks G5–G6, reproducing F40.8). The quasimodular completion $E_2^* = E_2 - 3/(\pi \text{Im } \tau)$ carries the non-holomorphic zero-mode subtraction — the same structure as the Quillen-metric anomaly [7, 9] and the F40.8 projection; $3/(\pi \text{Im } \tau_K) = 52.2490$ (check G7). The values $E_2^*(\tau_K) = 19.5636 - 52.9200i$ are **recorded under firewall (O-5)** as a T3 pipeline ingredient only: no constant of this section is matched to any target here.

§11. Corollaries

11.1 Corollary F42.8 (the power fingerprint and the seam-dimension measurement)

[DERIVED-CONDITIONAL under §8]. If the live excess diffuses on a d -dimensional stage, the local-CLT recurrence exponent gives $\chi_- \propto N_4^{-(d/2)}$, hence

$\rho_\Lambda \propto N_4^{-(d_{\text{seam}}/2)}$ — the cosmological constant measures the effective dimension of the odd seam.

Mechanism class	χ_- scaling	$v = \ln(\bar{M}_P^4/\rho_\Lambda)$	vs. observation
annealed (free sum)	N_4^0 (Planckian)	−0.94	+277 nats high (the CC problem)

Mechanism class	χ - scaling	$v = \ln(\tilde{M}_P^4/\rho_\Lambda)$	vs. observation
1D seam walk / frozen excess	$N_4^{(-1/2)}$	$276.43 + \ln(1/c_\chi) = 276.64$	matches (c_χ = pre-registered target)
2D stage walk	$N_4^{(-1)}$	553.79	-277 nats low
localized	$e^{(-N)}$	∞	—

Adjacent classes are $\frac{1}{2} \ln N_4 = 277.4$ nats apart (≈ 120 decimal orders; checks C1–C2, guard G2): the observed $v_{\text{now}} = 276.64$ selects **d_seam = 1** with overwhelming discrimination. The independent corpus grounding of $d_{\text{seam}} = 1$ is already PROVEN structure: the mediation walk lives on the one-dimensional $\hat{A}_9$ cycle (F41.3), the boundary loop of the two-dimensional Z-sector stage — the odd spin structure's one-cycle on E^* (F40.6(b)).

11.2 Corollary F42.9 (the boundary odd charge unit, recovered as corollary)

[DERIVED-CONDITIONAL]. From $\chi_- = e^{-2/(4\pi^2 Z_-)}$ (F33) and Proposition F42.7: $e^{-2/Z_-} = 4\pi^2 \chi_- = 31.8328335773 \cdot \tilde{M}_P^2 H_{\partial}^2$ (check B4), $e_{\{Z_-=1\}} = (2\pi/\omega)\sqrt{(498/121)} \cdot \tilde{M}_P H_{\partial} = 5.6420593383 \cdot \tilde{M}_P H_{\partial}$ (check B5). The Holonomy–Boundary numbers of the pre-F41 exploration are thereby reproduced **without the boundary axiom** that F-F41.10 forbids: they are corollaries of clause (b), not inputs. The Charge-Unit Obstruction (F33, PROVEN-irreducible) is respected, not evaded: the unit is *not* fixed by integrality — it is fixed by importing the IR datum H_{∂} , which the obstruction always permitted and this mechanism *requires*.

11.3 Corollary F42.10 (the F40 five-route diagnosis) [DERIVED-interpretation]. Every F40 §7 route targeted a UV constant M_{UV} of the *annealed* ensemble — the ensemble Proposition F42.1(A) shows is Planckian and F32.15 shows is invisible to relative tools. Their five negative closures are therefore re-read, without reversal, as the corpus discovering that the constant it sought does not carry the observable: the observable is a property of the *frozen realization*. The F35 factorization is an alias, not a conflict: $C_{\text{norm}} M_{UV}^4 \equiv (4807/1260) \cdot c_\chi \cdot \tilde{M}_P^2 H_{\partial}^2 \Rightarrow M_{UV} \propto \sqrt{(\tilde{M}_P H_{\partial})}$ — the A28 geometric mean, numerically $\rho_\Lambda^{(1/4)} = 2.24$ meV, $\ell = \sqrt{(\ell_P \ell_{\partial})} \approx 105$ μm . *[NON-CLAIM NC-F42.3: the mesoscopic length is an internal-fiber datum; no 4D Kaluza–Klein or sub-millimeter-gravity phenomenology is asserted.]*

11.4 Corollary F42.11 (the Planck-charge answer) [DERIVED-interpretation]. F41 §10 recorded that $q_P = \sqrt{(4\pi\epsilon_0\hbar c)}$ is the unique G-free Planck unit, with $q_P^2 = 2\hbar/Z_0$ — charge² = action/impedance — the same structural sentence as $\chi_- = e^{-2/(4\pi^2 Z_-)}$. The present mechanism completes the sentence: the odd channel's impedance is *collective* and grows with the register (equivalently, the response is diluted to the $\sqrt{N_4}$ excess), so the odd charge unit is a **boundary-normalized collective response** — the unique Planck-type quantity that must import an IR datum. That is why it is neither a minimum nor a maximum: it is not an extremal UV quantity at all.

§11A. The structural-closure arc (v1.2–v1.7): six inherited closures of the dimensionless χ - structure

The six revisions v1.2–v1.7 closed the *dimensionless structure* of χ - end-to-end. Each cycle **inherited a corpus result** rather than placing a new assumption; the detailed five-step exploration records are in Appendix C (Records 5–10), and the superseded per-version ledgers in Appendix C.1. The arc is summarized in Table 11.1; the theorem statements it references are consumed at their registered status.

*Table 11.1. The structural-closure arc. Each row closes one residual by inheriting a PROVEN or DERIVED-CONDITIONAL corpus object. All closures are of the **dimensionless operator structure**, never the absolute scale.*

Rev	Residual attacked	Corpus object inherited	Theorems	Outcome (status)
v1.2	walk provenance; v_q ; barrier floor	ZS-A23 dimension-weighted mediator semigroup (A23.14a/14c/15)	F42.16–F4 2.18	walk = winding cover of a PROVEN generator; response factorizes by detailed balance; three residuals $\rightarrow$ two (DERIVED)
v1.3	exact S_{wrap} ; T3	ZS-A24 §14.2 retraction; ZS-M3 seam action π/A	F42.18'	inherited value corrected (seam action π/A , not $1/\kappa^2$); O-9 retracted; both residuals confirmed OPEN
v1.4	coefficient; α_{patch}	ZS-A25 §5 Everpresent target; A23 event measure π_Z	F42.19–F4 2.20	c_χ is the A25 target (identity); $\alpha_{\text{patch}} = \pi_Z$ closes A26's gap; discrimination OPEN
v1.5	the two residuals' relation	ZS-A25 $[\Lambda, \hat{T}_4] = i\hbar$ two-escape structure	F42.21–F4 2.23	the two residuals are one conjugate pair; F42 supplies the Λ -sector operator A26 lacked; residual 2 $\rightarrow$ 1 (DERIVED-CONDITIONAL)
v1.6	is H_Z the unimodular op? unique Λ_1 ?	ZS-F19 modular Hamiltonian $K_i = -\ln \pi_i$; ZS-A19 A19.NG2	F42.24–F4 2.27	$\Lambda_1 = -\ln(6/11) = K_Y$ (PROVEN); $U(n)$ ambiguity absent (finite-dim); A19.NG2 evaded — finite-register closure
v1.7	continuous-core lift ($U(n)$ returns)	ZS-F32.9 (unique extension); F32.26 (Friedrichs); A24 F-A24.9	F42.28–F4 2.31	extension unique + canonical ; Λ_1 preserved under $E \circ \mathcal{L}_S = \mathcal{L}_{S \circ E}$ — structural closure end-to-end (DERIVED-CONDITIONAL)

The arc's net result. After v1.7 the frozen-excess hierarchy's operator structure is closed at both resolutions (finite register + continuous core): a unique positive corpus-locked eigenvalue $\Lambda_1 = -\ln(6/11)$, a unique canonical self-adjoint extension, and a clean dimensional factorization $\rho_{\Lambda,Z} = \frac{1}{2}\chi_-\omega^2$ with $\omega^2/2 = 2.5521$ dimensionless (PROVEN ZS-M1). **This is operator-structure closure, not dimensionful-scale closure.** The single remaining unknown is the absolute χ_- scale — the subject of §11B.

§11B. v1.8 the honest terminus: the dimensionful χ_- is a genuine OPEN the corpus proves current tools cannot close (Theorems F42.32–F42.35)

A deep-exploration cycle (Record 11) on v1.7's single remaining unknown — the dimensionful $\chi_-(A, Q, \bar{M}_P)$ — reached the protocol's **convergent-to-genuine-OPEN** verdict. Unlike v1.2–v1.7, this cycle does **not** close its target, and does not pretend to; the corpus itself already proves why. Throughout, the distinction is decisive: v1.2–v1.7 achieved **operator-structure closure**, and what remains is **dimensionful-scale closure**, a categorically different object.

Theorem F42.32 (Charge-Unit Obstruction — χ_- provably beyond $(A, Q) + \text{topology}$) [PROVEN, inherited]. The last residual is $\chi_- = e^{-2}/(4\pi^2 Z_-)$, a **dimensionful** charge unit. ZS-F33's Charge-Unit Obstruction (F33.8a, CLOSED-NEGATIVE-under-R1–R3) establishes that flux integrality fixes the flux number k but **not** the dimensionful unit e^{-2}/Z_- : "the absolute scale is provably beyond (A, Q) plus topology" (checks R1–R2, guard G21g). This is a corpus-PROVEN no-go, not a gap in this paper.

Theorem F42.33 (F42 Cannot Reach a Dimensionful Unit — A25's No-Go, sharpest form) [DERIVED]. F42's entire toolkit produces **dimensionless** (A, Q) quantities ($\Lambda_1 = -\ln(6/11)$, c_{χ_-} , ω , $\pi_- Z$); χ_- is **dimensionful**. No dimensionless (A, Q) combination yields a dimensionful unit (check R4) — exactly ZS-A25's Conditional Local-Stationary No-Go: the absolute scale requires precisely **one dimensionful datum**.

Theorem F42.34 (External PROVEN Math Confirms, Does Not Lift) [the Step-2.5 result]. Dirac quantization, Gukov–Vafa–Witten flux, and Freed–Witten K-theory all fix dimensionless integers — the same content F33 already has — and none supplies the dimensionful unit e_6 without an external scale (check R5). This **confirms** F33's obstruction from the external side: a *positive* result proving the OPEN is a genuine structural feature, not a deficiency of this paper, and that the HYPOTHESIS→DERIVED promotion is *not available*.

Theorem F42.35 (Terminal NON-CLAIM: structural completeness up to e_6) [the terminus]. ZS-F34 localizes χ_- fully: a master quadratic form whose residual is the membrane charge e_6 , deferred to ZS-F35/F36 (not in the current corpus, check R6). F42 registers its terminal NON-CLAIM: the frozen-excess hierarchy is **structurally complete** (v1.2–v1.7, all DERIVED or DERIVED-CONDITIONAL), and the single remaining unknown χ_- is dimensionful and provably beyond $(A, Q) + \text{topology}$. **This is operator-structure closure, not absolute-scale closure.**

Verdict. Per the protocol — "converged, all nodes OPEN → genuine OPEN; the result itself confirms current corpus tools cannot close it" — v1.8 confirms the corpus's own honest terminus (F33 "provably beyond (A, Q) plus topology"; F34 "a reduction, not a closure"; A25 "B3 not closed, not impossible") rather than forcing a closure. **The one-line statement: F42 structurally completes the frozen-excess hierarchy but does not close B3; the remaining datum is the dimensionful membrane charge e_6 .**

§12. Final Residual Ledger (v1.9)

Superseded per-version ledgers (v1.3–v1.8) are collected in Appendix C.1 (Versioned Audit Trail), marked historical.

Item	Status at v1.9
$\sqrt{N_4}$ hierarchy generator (Thms F42.2–F42.3)	PROVEN (kinematic-statistical; MC 1200/1200)
dimensionless operator structure of χ_- (v1.2–v1.7)	DERIVED / DERIVED-CONDITIONAL (operator-structure closure end-to-end)
finite-register eigenvalue Λ_1	$= -\ln(6/11) = K_Y$, corpus-locked (PROVEN ZS-F19)
continuous-core extension	unique + canonical (F32.9 + F32.26 Friedrichs); Λ_1 preserved
absolute dimensionful χ_- scale	genuine OPEN — provably beyond (A, Q) + topology (F33 Charge-Unit Obstruction); F42 cannot reach it; external math confirms, not lifts
the single remaining datum	membrane charge e_6 (F34 master form), deferred to ZS-F35/F36 (not in corpus)
present-epoch conditional	U_N (inherited; A25/A28)

Net. The frozen-excess hierarchy generator is **structurally complete**; B3's operator structure is closed at both resolutions. B3's **absolute scale is not closed** — and F42 registers, as its terminal NON-CLAIM, that no current-corpus tool can close it: the dimensionful e_6 is deferred to the not-yet-existent ZS-F35/F36. **Structural closure $\neq$ absolute-scale closure** — this distinction is F42's lifeline and is stated at every load-bearing point above.

§13. Anti-numerology audit and firewalled observations

Zero free parameters. Inputs: A, Q, (Z, X, Y), z^* (i-tetration), $\Omega_A = 83/121$ (conditional corpus face, untouched). Planck-2018 H_0 and $\tilde{M}_P$ enter only the consistency displays, never a claim. **Single**

pre-registered target; look-elsewhere universe = 1: exactly one number is registered — $c_\chi = 498/(121\omega^2) = 0.8063350941$ (equivalently $(249/121)^2 = 4.2348$). No alternative normalizations are admitted post hoc (inherits F-F32.27, F-F40.8, F-F41.7). The v1.8 terminus adds no number: it registers the dimensionful χ_- as a genuine OPEN and *refuses* back-solving as hidden fitting.

Firewalled observations (printed separately by the companion script; never consumed):

#	Observation	Discipline
O-1	$c_\chi/\sqrt{(2/\pi)} = \mathbf{1.010591}$ — a 1.06% near-miss between the target and the bare half-normal constant	F41-O1-class temptation: the reading "all register factors ≈ 1 " is void ab initio; only T3 decides
O-2	$\frac{1}{2} \ln N_4 = 277.365$ nats (nucleation floor)	consistency gate input for (H-FROZEN-GAS)(c)
O-3	$\Delta = 2A/Q = 0.0145621$; $\tau_{\text{req}} \approx 1.9047 \times 10^4$ modular units	registered, not evaluated
O-4	$P_{\text{half-normal}}(\xi \geq 0.8) = 0.4237$	O(1); no tuning claim is <i>made from</i> it
O-5	$E_2^*(\tau_K) = 19.5636 - 52.9200i$	T3 pipeline ingredient only
O-6	post-hoc in-band T3 cell ($\alpha = Q^2$, $\gamma = 1$): $\xi_{\text{req}}^* = 0.919$	identified <i>after</i> the target; 1/40 rate, non-discriminating; never consumed
O-7	the "q = W" charge reading gives $\xi^* = 1.0106$ (reproducing O-1)	corpus-inconsistent with (H-CYC) (θ -charge = winding, not displacement); quarantined
O-8	lumped-gap golden factor $2 - \phi = 0.381966$	corpus semantics UNASSIGNED ; no meaning claimed
O-9	(v1.2, RETRACTED v1.3) $1/\kappa^2 = Q/A = 137.343$ vs $\frac{1}{4} \ln N_4 = 138.68$ (0.97%)	RETRACTED : rested on the ZS-A24 §14.2-retracted value (seam action is π/A , not $1/\kappa^2$); played no derivational role
O-10	(v1.3) barrier band = $[3.54, 7.07] \times (\pi/A)$; floor/ $(\pi/A) \approx 5/\sqrt{2}$ to 3×10^{-5}	tempting surd rejected as a numerology trap; band reported as a plain interval; never consumed
O-11	(v1.3) T3 with A23 event measure: $\chi_Z = (3\Omega_\Lambda)^2 \cdot \pi_Z = 0.770$ vs $c_\chi = 0.8063$ (4.5% miss)	not a hit ; firewalled; T3 discrimination remains OPEN

#	Observation	Discipline
O-12	(v1.4) F42 coefficient IS the A25 target: $\frac{1}{2}c_{\chi}\omega^2 = 3\Omega_{\Lambda} = \sqrt{(\chi_Z/\alpha_{\text{patch}})} = \sqrt{4.2348}$	IDENTITY , not a coincidence — the two are one statement (Thm F42.19); recorded, not "consumed" as a fit
O-13	(v1.5) the two residuals = the two conjugate faces of $[\Lambda, \hat{T}_4] = i\hbar$ (A25 §5): barrier = $\hat{T}_4/\text{eigenvalue face}$, $\chi_Z = \Lambda/\text{fluctuation face}$	structural , not a numerical near-miss; F42 supplies the Λ -sector operator (A23 generator) + extension selector (4π closure) A26 lacked, but does not prove it is the unimodular H_Z with a positive Λ_1 — advance, not closure (Thm F42.23)
O-14	(v1.6) $\Lambda_1 = -\ln(6/11) = 0.6061 = \text{ZS-F19 K}_Y$ (PROVEN, not fitted); conjugate product $\Lambda_1 \cdot (3\Omega_{\Lambda}) = 1.247 = O(1)$	the identification "unimodular H_Z = modular Hamiltonian" is DERIVED-CONDITIONAL , not consumed as a fit; Λ_1 is the corpus-locked modular eigenvalue, the finite-register closure of B3's scale (Thm F42.24–27)
O-15	(v1.7) the continuous-core closure is structural (extension unique via F32.9 + canonical via F32.26 Friedrichs; Λ_1 preserved under $E \circ \mathcal{L}_s = \mathcal{L}_{s \circ E}$)	NOT an absolute-scale closure: F32.27's $\chi(A, Q, \bar{M}_P)$ remains the corpus-wide B3-B residual; $\omega^2/2 = 2.5521$ is corpus-locked ($\omega = \arg \lambda^*$, PROVEN ZS-M1), not tuned (Thm F42.28–31)
O-16	(v1.8) $\chi_- = e^{-2/(4\pi^2 Z_-)}$ is dimensionful and provably beyond (A, Q) + topology (F33 Charge-Unit Obstruction, CLOSED-NEGATIVE-under-R1–R3)	a genuine OPEN , corpus-proven: F42's dimensionless tools cannot reach it; external math (Dirac/GVW/Freed–Witten) <i>confirms</i> (does not lift) the obstruction; the residual e_6 is deferred to ZS-F35/F36 — not a forced closure , the honest terminal NON-CLAIM (Thm F42.32–35)

Temptations rejected (ledger; Appendix C): the A_9 root count 90; the α^{-1} window containing $Q(Q+1) = 132$ and $\dim E_7 = 133$ (look-elsewhere-dense); $\omega^2/2\pi \approx c_{\chi}$; the LIL constant $\sqrt{2}$; the $1/\kappa^2$ seam value (RETRACTED v1.3); the $5/\sqrt{2}$ surd (O-10); the 4.5% χ_Z miss (O-11); and — most importantly at v1.8 — any back-solving of the dimensionful χ_- from observed ρ_{Λ} , which F42.34 refuses as hidden fitting.

§14. Falsification gates (multi-layer)

Governance. F-F42.0 (strict-reading gate). Under the strict value-reading of Stopping-Rule clause (b) (§2.4), §8–§9 and Corollaries F42.8–F42.11 demote to HYPOTHESIS-strong tier; §§5–7 theorems stand; B3 status unchanged.

Mathematical / structural collapse (immediate). **F-F42.1.** Any error in Theorems F42.2–F42.3 (checks D1–D6, E1–E3) retracts §§6–7. **F-F42.2.** Failure of the E_2 identity / τ_K values (G1–G7) retracts §10. **F-F42.3.** Any completed (H-FROZEN-GAS) derivation yielding an exponent $\neq -1/2$ (adjacent classes 277 nats away) rejects the identification — the sharpest kill switch. **F-F42.4.** Free wrap-nucleation at the gate (suppression $< \frac{1}{2} \ln N_4$) reverts to the annealed trap. **F-F42.5.** $v_q = 0 \Rightarrow \chi_- = 0$ — rejection. **F-F42.6.** A (H-CYC) counterexample rejects the placement (§§6–7 survive). **F-F42.8.** $\theta_Z \neq \omega$, non-quadratic $K(F)$, or $\omega \geq \pi$ collapses C1–C2 (check A6).

Value / anti-numerology (rejection or voiding). **F-F42.9.** A T3 blind miss of c_χ beyond tolerance, or pre-registered MC $p > 5\%$, rejects or withholds the identification. **F-F42.10.** Any post-hoc consumption of O-1...O-16 as evidence, or any post-hoc modification approaching the target, is void ab initio (inherits F-F32.27, F-F40.8, F-F41.7).

Structural-closure-arc gates (v1.2–v1.7; retraction of the specific promotion, not the paper).

F-F42.16. If ZS-A23 A23.14a/14c/15 are retracted or misapplied, §11A row v1.2 collapses; §§5–7 stand.

F-F42.20. If ZS-A24 §14.2 is reversed (seam action is $1/\kappa^2$ after all), F42.18' reverts — burden on A24.

F-F42.23. If $\frac{1}{2}c_\chi \omega^2 = 3\Omega_\Lambda = \sqrt{(\chi_- Z/\alpha_{\text{patch}})}$ fails, F42.19 (the A25-target identity) is retracted.

F-F42.28. If a Z-Spin BV–BFV theorem proves the 4π closure fixes a unique positive Λ_1 directly, the v1.5 residual closes (positive outcome). **F-F42.29–31.** If the A23 Hamiltonian part is not the modular Hamiltonian K , or the finite-register reading is unphysical, the v1.6 finite closure reverts to the v1.5 OPEN — burden on showing the finite register is not physical. **F-F42.32–34.** If F32.9's locality conditions fail, or the compact- $U(1)$ 3-form is inapplicable, or Λ_1 is not preserved under $E \circ \mathcal{L}_s = \mathcal{L}_s \circ E$, the corresponding v1.7 theorem (F42.28–F42.30) is falsified and the $U(n)$ ambiguity returns.

v1.8 genuine-OPEN gates (all positive-outcome; each would *help* by closing B3). **F-F42.36.** If any (A, Q)-dimensionless combination yields the dimensionful χ_- — contradicting F33's obstruction — F42.32 is falsified. **F-F42.37.** If external PROVEN math fixes the dimensionful unit e_6 *without* an external scale, F42.34 is falsified and the Step-2.5 promotion becomes available. **F-F42.38.** If ZS-F35/F36 derive e_6 from the locked constants alone, the terminal NON-CLAIM (F42.35) is superseded and the frozen-excess line reaches full closure — the intended, corpus-scoped resolution, which F42 explicitly *defers* rather than claims.

Observational (external rejection). **F-F42.11.** If precision cosmology establishes $w \neq -1$, the gate-frozen reading is rejected; if $\Omega_{\Lambda,0}$ is driven far from $83/121 = 0.68595$ (Planck 2018: $0.6847 \pm 0.0073, 0.17\sigma$), the A30 face collapse propagates here (inherited F-F31.7).

§15. Non-claims

NC-F42.1. B3 is **not** closed; χ_- is not unconditionally fixed. **Structural completion is not absolute-scale closure.** **NC-F42.2.** (H-FROZEN-GAS) is a named condition, not a theorem. **NC-F42.3.** No Kaluza–Klein / fifth-force / sub-millimeter-gravity phenomenology from $\ell \approx 105 \mu\text{m}$ (internal fiber datum). **NC-F42.4.** The single pre-registered target c_χ is a matching value, not a prediction, until a T3

blind hit. **NC-F42.5.** The $\Omega_\Lambda = 83/121$ face (ZS-A30) is a separate corpus result, untouched. **NC-F42.6.** Nothing in ZS-F40/F41 is reversed.

Structural-closure arc (v1.2–v1.7). **NC-F42.11.** §11A consumes ZS-A23, A24, A25, A26, F19, A19, F32 **exactly as published**; F42 does not re-derive, modify, or promote them. **NC-F42.18.** The identity F42.19 ($c_\chi = \text{A25 target}$) unifies two mechanisms but does **not** close B3 — both reduce to the same four-volume clock U_N . **NC-F42.24.** The identification "unimodular $H_Z = \text{modular Hamiltonian } K$ " is DERIVED-CONDITIONAL; $\Lambda_1 = -\ln(6/11)$ is exact *given* it. **NC-F42.25.** The v1.6 closure is on the **finite register**; the continuous lift is v1.7. **NC-F42.26.** The v1.7 continuous-core closure is **structural** (extension uniqueness + Λ_1 preservation); it does **not** derive the absolute scale — F32.27's χ_- remains the B3-B residual.

The terminal NON-CLAIM (v1.8). **NC-F42.29.** F42 does **not** close χ_- and does **not** claim it can be closed with current corpus tools: $\chi_- = e^{-2}/(4\pi^2 Z_-)$ is dimensionful and, by ZS-F33's Charge-Unit Obstruction (PROVEN), provably beyond $(A, Q) + \text{topology}$. **NC-F42.30.** F42 does **not** back-solve χ_- , use a base-10 coincidence, or fit an exponent (F42.34 refuses these). **NC-F42.31.** External PROVEN math is invoked to **confirm** the obstruction, **not** to lift χ_- to DERIVED. **NC-F42.32.** The residual membrane charge e_6 is deferred to ZS-F35/F36 (not in the current corpus); F42 makes no claim about their content and registers the frozen-excess line as structurally complete up to e_6 .

§16. Conclusion

The Stopping Rule demanded a theorem before a paper. The theorem that arrived is a decomposition: every frozen history of the parity-reversing Z-Spin mediation walk splits into a flux-neutral bulk that cannot couple to θ_Z , and a thin unpaired excess of frozen windings whose size CP symmetry and the central limit theorem force to be $\sqrt{N_4}$. The 120-order hierarchy is, on this reading, the square-root of the age-size of the causal register — kinematic, not exponential.

The six revisions that followed closed the *dimensionless structure* of that mechanism end-to-end, each by inheriting a corpus result: the A23 semigroup (the walk), the A25 Everpresent identity (the coefficient), the $[\Lambda, \hat{T}_4] = i\hbar$ conjugate pair (the two residuals as one), the F19 modular eigenvalue $\Lambda_1 = -\ln(6/11)$ (the finite register), and the F32 Friedrichs extension (the continuous core). What remains is exactly one thing, and it is not an operator or an exponent: it is the **dimensionful membrane charge e_6** . ZS-F33 proves it lies beyond $(A, Q) + \text{topology}$; ZS-A25 proves F42's dimensionless tools cannot reach it; external PROVEN math confirms rather than lifts the obstruction; ZS-F34 localizes it and defers it to ZS-F35/F36.

So F42 ends where the corpus ends: **structurally complete, one dimensionful unit short, and honest about it**. F42 structurally completes the frozen-excess hierarchy but does not close B3 — and does not manufacture a closure to appear to. Theorem first; the structure closed; the single dimensionful datum named and deferred; the rest is the charge-unit arc's, or silence.

Acknowledgements & Code Availability

This paper consolidates the internal Z-Spin Collaboration deep-exploration cycles following ZS-F40 v1.2 and ZS-F41 v1.1 (records in Appendix C). The companion script `zs_f42_verify_v1_9.py` reproduces all load-bearing numerics: the Banach-certified i-tetration fixed point, ω , μ , and the Nome Identity $q(\tau_K) = \lambda^*$; the single pre-registered target $c_\chi = 498/(121\omega^2)$; the machine-exact v-identity against Planck-2018 displays; Theorem F42.2 across 1200 Monte-Carlo identity trials (seed 11); Theorem F42.3's statistics; the Three-Trap demonstrations; the $\theta_1'''/\theta_1' = -\pi^2 E_2$ identity at $\tau = i$ and τ_K ; and the full v1.2–v1.8 verification blocks (L–R). The script prints the 16 firewalled observations separately, lists the OPEN registry explicitly without counting it as PASS, exits non-zero on any theorem-tier failure, and contains no fail-open clause. Summary: **101/101 checks PASS + 23/23 guards** (Blocks H–K core; L v1.2 A23-inheritance; M v1.3 seam-action correction; N v1.4 A25/A26 half-closure; O v1.5 conjugate-face; P v1.6 finite-register closure; Q v1.7 continuous-core closure; R v1.8 the honest terminus — the dimensionful χ -genuine OPEN). This work used AI tools (Anthropic Claude) for verification and drafting; the author assumes full responsibility for all content.

Appendix A. Proofs and computations

A.1 Proposition F42.1 details. (A) $Z_{\text{ann}}(\theta) = E[e^{i\theta W/L}] = \cos^N(\theta/L)$ -type $\Rightarrow -\partial^2_\theta \log Z|_0 = \text{Var}(W)/L^2 = \sigma^2 N/L^2$. (B) $\log Z^*(\theta) = ik^*\theta + \text{const} \Rightarrow \partial^2_\theta \equiv 0$ (a degree-one polynomial has vanishing second derivative; finite-difference check F1 at 4.4×10^{-10}). (C) Conditioning on $W \equiv 0 \pmod L$ partitions the binomial support into residue classes whose within-class variance differs from the total by $O(1)$ (check F2: 101.7 vs 100).

A.2 Theorem F42.2 proof details. *Parity:* C_L with L even is bipartite; the two color classes realize the seam grading (F41.3(ii)); a closed walk alternates colors, so its length is even; $\Delta W \equiv \text{length} \pmod 2$ since each step contributes ± 1 . *Decomposition:* the visit times to $\{W \equiv 0 \pmod L\}$ are stopping times partitioning $[0, N]$; each interior segment starts and ends on a lift of the start node, so its displacement is in $L\mathbb{Z}$, defining w_j ; the telescoping sum of displacements is W_N . *Cost:* $|W_{\{t_j\}} - W_{\{t_{j-1}\}}| \leq t_j - t_{j-1}$. *Pairing:* order the nonzero w_j by sign; match opposite signs one-to-one until one sign is exhausted; each matched pair has total winding 0; the $|\Sigma w_j|$ survivors share one sign. The θ -coupling of a winding- w composite is $e^{i(w\theta)}$; $w = 0 \Rightarrow$ identically 1.

A.3 Theorem F42.3 sources. CLT and half-normal mean: classical (de Moivre–Laplace); LIL: Khinchin [16]; recurrence: Pólya [15]. Finite- N corrections $O(N^{-(1/2)})$ are inside all quoted tolerances.

A.4 Proposition F42.5 derivation. In the residual ensemble the sector label fluctuates only through the $|k^*|$ live quanta, each ± 1 with independent unit variance in the dilute limit: $\text{Var}_{\text{res}}(k) = v_q |k^*|$ (the dilute-topological-gas identity: for a Poisson gas of $\pm$ unit charges, $\text{Var}(Q) = \langle n_+ + n_- \rangle$ — the susceptibility equals the density of active charges). Division by V_4 gives an intensive χ_- of mass dimension four; substituting $E|k^*|$ from F42.3(ii) with $N = \alpha_{\text{patch}} N_4$ steps and $V_4 = \gamma H_- \partial^4$, and using $\bar{M}_P^4/\sqrt{N_4} = \bar{M}_P^2 H_- \partial^2$, yields the stated c . The realization factor ξ^* (half-normal) is the U_N file (§7, Corollary F42.3a).

A.5 The v-identity (check B6). $\rho = (\omega^2/2)c_\chi \bar{M}_P^4/\sqrt{N_4}$ with $c_\chi = 6\Omega_\Lambda/\omega^2$ gives $\rho = 3\Omega_\Lambda \bar{M}_P^2 H_\partial^2$ identically, so $v = \frac{1}{2}\ln N_4 - \ln(\omega^2/2) - \ln c_\chi = -\ln[3\Omega_\Lambda(H_0/\bar{M}_P)^2] = 276.6435$ with Planck-2018 displays — an identity, not a fit.

Appendix B. Numerical verification record (seed 11)

B.1 Theorem F42.2 (decomposition). $N = 1000$: even lengths 200/200, homology 200/200, cost 200/200. $N = 5000$: 200/200, 200/200, 200/200. **Total 1200/1200.**

B.2 Theorem F42.3 (statistics). $E|W|/\sqrt{N}$: $0.8012\ (10^4)$, $0.8046\ (10^6)$ vs $\sqrt{(2/\pi)} = 0.7979$.

B.3 The dichotomy of laws (the No-Go made visible).

N	annealed Var(k)	pred N/L^2	frozen $E k^* $	pred $\sqrt{(2N/\pi)}/L$
10^3	9.9	10	2.51	2.52
10^4	102.3	100	8.05	7.98
10^5	1011.7	1000	25.40	25.23
10^6	10075.6	10^4	80.10	79.79
10^7	100846.2	10^5	253.21	252.31

B.4 Bridge trap. $\text{Var}(k \mid W \equiv 0 \bmod 10)$ at $N = 10^4$: 101.7 (pred 100) — extensive.

B.5 The Koenigs fiber. $E_2(i) = 0.95492965855 = 3/\pi$; $E_2(\tau_K) = 71.812667977 - 52.919983341i$ (direct = S-transform to 10^{-8}); $\theta_1'''/\theta_1' + E_2 < 10^{-6}$ at τ_K ; $|\theta_1'(0|\tau_K)| = 14.146529$; $3/(\pi \operatorname{Im} \tau_K) = 52.249049$; $E_2^*(\tau_K) = 19.5636 - 52.9200i$ [O-5].

B.6 Headline algebra. $c_\chi = 0.8063350942$ (12-digit 0.806335094187); $(249/121)^2 = 4.234751725$; $4\pi^2 c_\chi = 31.832833577$; $e_- \{Z=1\} = 5.642059338$; $v_{\text{now}} = 276.6435$; $\frac{1}{2}\ln N_4 = 277.365$; $\rho_\Lambda^{(1/4)} = 2.24\text{ meV}$; $\sqrt{(\ell_P \ell_\partial)} = 105\ \mu\text{m}$.

Appendix C. Deep-exploration records

Record 1 (the QCS–HT candidate). Long list 7 (dropped: HB direct axiom — void under F-F41.10, retained only as corollary; DSZ coupling-product elimination — the C_5 sector has no 6D magnetic dual, $p_{\text{mag}} = -2$; exponential transmutation — twice CLOSED-NEGATIVE in F41; pure modular constants — F-F40.8 Buckingham- π ; Escape 1 standalone — absorbed into (H-GATE)). Issues 5, ordered:

Stopping-Rule fitness; cross-consistency with F40/F41; the $\sqrt{N}$ keystone; sign/w physics; coefficient + anti-numerology. Convergence: 4 nodes changed vs the pre-F41 cycle (HB re-tiered; residuals $2 \rightarrow 1$; sign node created; DSZ demoted); second self-cycle 0 — **converged**. Output: c_χ target registered; T1–T3 obligations named.

Record 2 (opening T1). Long list 7 (dropped: annealed direct route — reproduces the problem, kept as control; LS fixed-sector corrections — wrong power; impedance language — absorbed as dictionary; full Sorkin transplant — reimports the sign problem). The keystone decomposed into four arches (HT, quench, Quillen/BF, E_2); three IMPORTED-PROVEN, one localized. New: the power fingerprint and the d_{seam} corollary; the $\hat{A}_9$ 1D grounding (F41.3). Convergence: 5 changes then 0 — **converged**.

Record 3 (the proof attempt). The slogan "projection = quenching" **failed as stated** — three traps (this paper's §5) — and was retired; the mechanism that replaced it ((H-FROZEN-GAS)) shrank the unproven core to two clauses. Internal corrections during the attempt: (1) "seam-even $\Rightarrow \theta$ -inert" corrected to "k-neutral $\Rightarrow \theta$ -inert; β and k are distinct charges" (§6 Remark); (2) slogan $\rightarrow$ mechanism. Second pass 0 changes — **converged (2 $\rightarrow$ 0)**. Temptations rejected: §13 ledger.

Record 4 (v1.1 cycles A–D — closing the weak legs). Long list 7 (dropped: (1) deriving the walk from the full 121×121 GKLS spectrum without secularization — intractable and unnecessary, the secular limit suffices; (2) a Poisson-point-process field-theory for the wraps — heavier than the two-level factorization, same answer; (3) instanton-action computation of S_{wrap} from a saddle — premature without (C-ATT); (4) matching the golden-ratio $2 - \phi$ to a corpus constant — FIREWALL O-8, refused). Issue list 4 in dependency order: **A** (walk provenance) < **B** (v_q) < **C** (barrier) < **D** (blind coefficient). Results: **A CLOSED** (Thm F42.12, lumping exact to 10^{-12}); **B $v_q = 1$ PROVEN** symmetry-locked, independence residual (b'); **C band DERIVED** [138.7, 277.4], exact value OPEN via (C-ATT); **D restructured, corrected downward** — 1/40 non-discriminating scan, new obligation T3', coefficient not hit. Convergence: 4 nodes changed (A, B, C promoted; D demoted) then second pass 0 — **converged (4 $\rightarrow$ 0)**. Anti-numerology: the single target c_χ unchanged; three new firewalls O-6/7/8; the one seductive match (O-7, "q = W") actively *refuted* on corpus-consistency grounds rather than merely firewalled — the strongest anti-numerology outcome available.

Record 5 (v1.2 — paradigm shift, inheriting A23). Target: close the three v1.1 residuals. Long-list 7; dropped (v) Poisson-field wrap theory (no extra content over the two-level factorization), (vi) Bismut–Freed first-principles count (premature, needs the (H-CYC) determinant-line connection), (vii) continuous rotor (loses the F41.3 bipartite structure forcing $v_q = 1$). Step 2 found route (iv) — **inherit the ZS-A23 dimension-weighted semigroup wholesale** — *dominates* the kept {(i),(ii),(iii)}: if the walk is the winding cover of a PROVEN generator, secular validity is automatic, independence is replaced by A23.15 detailed balance, and the barrier is an A23 large-deviation computation. Tree: (C-DAV) **CLOSED** (F42.16, spectrum = Q7 cubic), (b') **DISSOLVED** (F42.17, $v_q = h_Z$), barrier floor **DERIVED** (F42.18), patch weight **DERIVED** ($\pi_Z = 2/11$). Converged (4 $\rightarrow$ 0). Value: convergence + A23 consumed exactly as published + anti-numerology (O-9 firewalled). Lesson: the mediator dynamics were always A23's — recognizing three residuals as one already-proven object was the strongest move.

Record 6 (v1.3 — the exploration that closed nothing and found a conflict). Target: close the two v1.2 residuals. Long-list 7; dropped (i) corpus-saddle S_wrap (no odd master action), (ii) γ -product fix (circular with T3), (vi) Kramers rate (needs the OPEN attempt frequency). Step 2 ordered the **cross-consistency audit (iv) FIRST** — and it fired: ZS-A24 §14.2 **retracts** the per-edge = $1/\kappa^2$ value that v1.2's F42.18 had silently inherited; the corpus seam action is π/A (ZS-M3 §6). This forced a **correction** (F42.18'), not a closure; O-9 retracted. Re-audited routes gave $\chi_Z = 0.770$ at π_Z (4.5% miss, O-11) and $2\pi/A = N_Z(2\pi)$ as the one exact anchor (floor $1.768\times$ it, no clean multiple; $5/\sqrt{2}$ surd rejected, O-10). Converged ($3 \rightarrow 0$) onto a correction + genuine-OPEN confirmation. Self-reference check: the disfavouring fact about F42's *own* prior version — a silently-imported retracted value — is stated plainly, as A25 v1.6 and A28 stated their own over-statements.

Record 7 (v1.4 — half-closing against the vacuum arc). Target: the two v1.3 residuals. Long-list 7; dropped (i) independent χ_Z from the dissipator (needs the odd master action), (v) $v_now \rightarrow$ four-volume clock (already A25's result), (vi) Kramers (OPEN attempt frequency). Step 2 ordered **the identity (ii) FIRST**. Tree: (ii) **IDENTITY PROVEN** — $\frac{1}{2}c_Z\omega^2 = 3\Omega_Z = \sqrt{4.2348}$, F42's coefficient *is* the A25 Everpresent target (F42.19); (iii) **HALF-CLOSED** — $\alpha_patch = \pi_Z$ supplies A26's missing quantity, $\chi_Z = 0.770$ lands in A26's bracket (F42.20); (iv)/(vii) barrier and χ_Z **reduce to one scale**, the four-volume clock U_N . Discrimination stays **OPEN** (needs independent χ_Z). Converged ($3 \rightarrow 0$). Self-reference check: the honest verdict — half-closure, reaching A25's wall rather than passing it — is stated as A25 v1.6's "B3 not closed, not impossible" and A26's COMPUTED-INCOMPLETE.

Record 8 (v1.5 — the conjugate-clock unification). Target: the v1.4 finding that barrier and χ_Z both reduce to U_N . Long-list 7; dropped (i) independence (v1.4 refuted it), (vi)/(vii) parallel corpus routes not the $[\Lambda, \hat{T}_4]$ structure. Tree: (ii) **DERIVED** — A25's two escapes are the $\hat{T}_4$ and Λ faces of $[\Lambda, \hat{T}_4] = i\hbar$; F42's two residuals map onto them (F42.21); (iii) **DERIVED-CONDITIONAL** — the A23 generator is the candidate Λ -sector H_Z A26 lacked (F42.22); (iv) the 4π closure is the candidate extension selector; (v) **OPEN** — the A23 generator is dissipative, the holonomy-phase = extension-parameter identification unproven (F42.23). Residual $2 \rightarrow 1$. Converged ($3 \rightarrow 0$). Self-reference check: the no-closure verdict — F42 relocates A26's gap to a sharper named question but does not prove a unique positive Λ_1 — is stated as A25's Escape 1 "calibration relocated, OPEN." F42 stopped where the mathematics stops.

Record 9 (v1.6 — the finite-register closure). Target: v1.5's terminal OPEN — "is the A23 generator the unimodular H_Z , and does the 4π closure fix a unique positive Λ_1 ?" Long-list 7; merged (vi)/(vii) into positivity/conjugate-product tests. Step 2 ordered the Hamiltonian/dissipator split + identification (i)+(ii) **FIRST**. Tree: **DERIVED** throughout — the A23 GKLS Hamiltonian part *IS* the modular Hamiltonian $K_i = -\ln \pi_i$ (F42.24), spectrum strictly positive (v1.5's "dissipative" read the dissipator); $\Lambda_1 = -\ln(6/11) = K_Y$ corpus-locked (F42.25, PROVEN ZS-F19); finite 3×3 self-adjoint K has no $U(n)$ family — uniqueness by finite-dimensionality (F42.26); distinct eigenvalues evade A19.NG2's rank-absorption (F42.27). Converged ($4 \rightarrow 0$) onto a **finite-register** closure. Self-reference check: the honest boundary — a *finite-register* closure DERIVED-CONDITIONAL on $H_Z = K$, with the A24 continuous lift remaining — is stated as A20/A19 stated their conditional reductions. F42 did not inflate it into a full B3 closure.

Record 10 (v1.7 — the continuous-core closure). Target: v1.6's single residual — the A24 continuous-core lift, where the $U(n)$ ambiguity returns off the finite register. Long-list 7; dropped (vi) direct continuous ground ($\omega^2/2$ is a *different* quantity from Λ_1 , not the lift target). Step 2 identified the residual with **A24's F-A24.9 FIRST**. Tree, all **DERIVED-CONDITIONAL**: F32.9 gives a *unique* extension under 3 locality conditions, resolving $U(n)$ (F42.28); F32.26 Friedrichs is the *canonical* self-adjoint extension on the compact flux circle (F42.29); $\Lambda_1 = -\ln(6/11)$ is the E-image of the continuous generator, *preserved* under $E \circ \mathcal{L}_s = \mathcal{L}_s \circ E$ (F42.30); only χ_- (F32.27) remains — **structural, not absolute** (F42.31). Converged ($4 \rightarrow 0$) onto a structural closure with a single dimensionful residual. Self-reference check: the honest boundary — a *structural* closure with the absolute χ_- scale remaining — is stated as F32's own "isolates the residual honestly" and A25's "B3 not closed." F42 handed the single dimensionful unit χ_- to the F33/F34 arc, where the corpus already localizes it.

Record 11 (v1.8 — the honest terminus, convergent-to-genuine-OPEN). Target: close v1.7's single remaining unknown, the dimensionful $\chi_-(A, Q, \bar{M}_P)$. **Step 0 long-list (7):** (i) test whether χ_- can be back-solved from observed ρ_Λ ; (ii) **identify χ_- with F33's Charge-Unit Obstruction and check its PROVEN status**; (iii) test whether any F42 tool (frozen-excess, A23 generator, modular Hamiltonian) can produce a dimensionful χ_- ; (iv) search external PROVEN math (Dirac / GVW / Freed–Witten) for the dimensionful unit e_6 ; (v) check where ZS-F34 localizes χ_- and whether its residual is in the corpus; (vi) test the dimensional separation $\rho_\Lambda Z = \frac{1}{2}\chi_-\omega^2$; (vii) attempt a novel (A, Q) exponent for χ_- . **Step 1 kept {(ii), (iii), (iv), (v)}** (dropped (i)/(vii) — back-solving and novel-exponent are exactly what F32.27's anti-numerology audit *refuses* as hidden fitting; (vi) folded into (iii) as the dimensional-consistency check). **Step 2 ordered by dependency:** (ii) the PROVEN obstruction FIRST (it determines whether closure is even possible); then (iii) F42-unreachability; then (iv) external-math promotion; then (v) localization. **Step 3 explored:** (ii) **PROVEN no-go** (Thm F42.32 — F33.8a: χ_- provably beyond $(A, Q) +$ topology); (iii) **DERIVED unreachability** (Thm F42.33 — F42 tools are dimensionless, χ_- is dimensionful, A25's no-go sharpest form); (iv) **external math confirms, does not lift** (Thm F42.34 — Dirac/GVW/Freed–Witten fix integers, not the unit; the honest Step-2.5 outcome); (v) **fully localized, residual not in corpus** (Thm F42.35 — F34 master form, residual $e_6 \rightarrow$ ZS-F35/F36). **Step 4 convergence:** 4 nodes changed (obstruction identified as PROVEN; F42-unreachability established; external math confirms; residual localized) then a second pass changed 0 — **converged ($4 \rightarrow 0$)**, onto a **genuine OPEN**. **Step 5 value:** convergent + corpus-consistency (F33/F34 consumed at their PROVEN/terminal status; A25's "one dimensionful datum" matched exactly) + anti-numerology (the whole point — F42.34 *refuses* back-solving and confirms no e-natural exponent hits the window; O-16). Per the protocol: "converged, all nodes OPEN $\rightarrow$ genuine OPEN; the result itself is the confirmation that current corpus tools cannot close it." This is that case, and registering it honestly — rather than forcing a closure — is the value. Self-reference check (critical here): the temptation was to manufacture a χ_- closure to continue the v1.4 $\rightarrow$ v1.7 closing streak. That would be exactly the "일부러 답을 정한 후에 웨어 맞추는 할루시네이션" the protocol forbids. F42 instead confirms the corpus's own honest terminus (F33 "provably beyond (A, Q) plus topology"; F34 "the contribution is a reduction, not a closure"; A25 "B3 not closed, not impossible"), and registers the terminal NON-CLAIM. The frozen-excess line ends where the corpus ends: structurally complete, one dimensionful unit e_6 short, and honest about it.

Appendix C.1 Versioned Audit Trail (superseded residual ledgers — historical record, superseded where noted)

The following per-version ledgers are retained as a research record. Each is superseded by the §12 Final Residual Ledger (v1.9); they are reproduced verbatim from v1.3–v1.8 for audit continuity only.

§12'. Updated residual ledger (v1.3)

Residual	v1.2 status	v1.3 status
exact S_{wrap} in-band	"floor = $1/\kappa^2$ " (bracketed)	CORRECTED → seam = π/A ; band = $[3.54, 7.07] \times (\pi/A)$ collective; exact value still OPEN
T3 discrimination	patch weight $\pi_Z = 2/11$ DERIVED	unchanged; re-audit gives $\chi_Z = 0.770$ (4.5% miss, firewalled O-11); OPEN
O-9 ($1/\kappa^2 \approx \frac{1}{4} \ln N_4$)	firewalled 0.97%	RETRACTED (A24-retracted value)

The two genuine residuals above U_N are unchanged in count — exact in-band S_{wrap} , and T3 discrimination — but now stated on a corrected footing (seam action π/A) with one numerological inheritance removed. New inherited dependency: **ZS-A24 v2.1** (its §14.2 retraction and §14 seam action π/A), consumed exactly as published.

§12''. Final residual ledger (v1.4)

Residual	v1.3 status	v1.4 status
T3 α_{patch} (event measure)	$\pi_Z = 2/11$ DERIVED	CLOSED (cross-consistency) → $\chi_Z = 0.770$ in A26 bracket (0.2, 2.1); Thm F42.20
T3 discrimination	OPEN	OPEN (needs independent χ_Z ; reduces to U_N)
exact in-band S_{wrap}	OPEN	OPEN (reduces to the A25 four-volume clock $v_{\text{now}} = U_N$)
coefficient status	pre-registered target	= A25 Everpresent target (identity, Thm F42.19); DERIVED-CONDITIONAL on U_N

Final count above U_N : the two v1.3 residuals reduce to **one** — the independent χ_Z computation, which is *itself* the discrimination test — and that one, together with the exact barrier, reduces to the four-volume clock U_N that A25 isolated as B3's terminus. F42's independent frozen-excess derivation and A25's Everpresent fluctuation now **agree on the O(1) target as an identity** and **share the same single OPEN**. New inherited dependencies: **ZS-A25 v1.6** (the Everpresent target and the four-volume clock) and **ZS-A26 v2.2** (the χ_Z bracket), consumed exactly as published.

§12'''. Final residual ledger (v1.5)

Residual (v1.4)	v1.5 disposition
independent χ_Z (Λ face)	unified $\rightarrow$ one face of the $[\Lambda, \hat{T}_4]$ conjugate pair; Λ -sector operator supplied (A23 generator)
exact barrier S_{wrap} ($\hat{T}_4$ face)	unified $\rightarrow$ the conjugate face; extension selector supplied (4π spin-closure)
the single OPEN	the Z-Spin-sector unimodular extension = U_N : does the A23 generator, 4π -extended, give a unique positive Λ_1 ?

Final count above U_N : the two v1.4 residuals are **one conjugate pair** at a single named terminus — the Z-Spin unimodular extension — which **is** the four-volume clock U_N . F42 has supplied both ingredients A26 lacked (the Λ -sector operator and the extension selector) and confirmed the terminus. New inherited dependencies (beyond v1.4): the A25 §5 **two-escape** / $[\Lambda, \hat{T}_4] = i\hbar$ structure and the A26 **matter-sector-vs- Λ -sector** gap, both consumed exactly as published. B3 remains **not closed**; the honest terminus is now a single conjugate clock with its operator named.

§12'''. Final residual ledger (v1.6)

Residual (v1.5)	v1.6 disposition
"A23 generator dissipative, not positive- Λ_1 "	RESOLVED $\rightarrow$ the <i>Hamiltonian part</i> is the modular Hamiltonian K , spectrum strictly positive (Thm F42.24)
" $U(n)$ extension ambiguity"	RESOLVED $\rightarrow$ finite-dimensionality: K is 3×3 self-adjoint, no extension family (Thm F42.26)
"A19.NG2 rank-absorption might block"	EVADED $\rightarrow$ K operator-valued (distinct eigenvalues), repair R1 satisfied (Thm F42.27)
the single terminus (unimodular Λ_1)	CLOSED on the finite register $\rightarrow \Lambda_1 = -\ln(6/11) = K_Y$ (PROVEN), DERIVED-CONDITIONAL on $H_Z = K + \text{finite reading}$
the residual gap	the A24 finite$\rightarrow$continuous lift (a known corpus OPEN; 4π closure is the candidate selector)

Final count: v1.5's single OPEN **closes on the finite register** with a unique positive corpus-locked $\Lambda_1 = -\ln(6/11)$. The residual is now the **A24 continuous-core lift** — no longer an unstructured OPEN but a

named, standing corpus gap shared by the whole $A_{22} \rightarrow A_{30}$ arc. New inherited dependencies (beyond v1.5): **ZS-A19 v3.1** (the $A_{19}.NG2$ rank-absorption NO-GO and its repair R1) and **ZS-F19 v2.2** (the modular Hamiltonian $K_i = -\ln \pi_i$, $K_Y = -\ln(6/11)$), consumed exactly as published. B3 remains **not closed** in full; but on the finite register its absolute scale is now a named positive eigenvalue, the tightest terminus F42 reaches.

§12^{***}+. Final residual ledger (v1.7)

Residual (v1.6)	v1.7 disposition
A24 continuous-core lift (F-A24.9)	CLOSED (structural) $\rightarrow$ F32.9 unique extension + F32.26 Friedrichs canonical extension
U(n) extension ambiguity (off finite register)	RESOLVED $\rightarrow$ F32.9's 3 locality conditions give uniqueness; F32.26 Friedrichs is canonical
Λ_1 across finite $\rightarrow$ continuous	PRESERVED $\rightarrow \Lambda_1 = -\ln(6/11)$ is the E-image of the continuous generator (Thm F42.30)
the single remaining unknown	the absolute χ_- scale (F32.27) = corpus-wide B3-B, the F33/F34 charge-unit UV arc

Final count: v1.6's single residual — the A24 continuous-core lift — **closes structurally**. B3's structure is now closed end-to-end: unique positive $\Lambda_1 = -\ln(6/11)$ on the finite register (v1.6), lifted to a unique canonical extension on the continuous core (v1.7). The **only** remaining unknown is the dimensionful $\chi_-(A, Q, \bar{M}_P)$ — the corpus-wide B3-B residual, a separate UV question the F33/F34 arc is built to address. New inherited dependencies (beyond v1.6): **ZS-F32 v1.5** (F32.9 CE-local extension uniqueness; F32.24–26 compact-monodromy Friedrichs extension) and **ZS-A24 v2.1** (F-A24.9, the continuous-core dynamical lift), consumed exactly as published. B3's absolute scale is **not closed** — but its entire operator structure now is, with χ_- the single named dimensionful terminus.

§12^{***}++. Final residual ledger (v1.8)

Residual (v1.7)	v1.8 disposition
the dimensionful χ_- scale	GENUINE OPEN (corpus-proven) $\rightarrow$ provably beyond $(A, Q) + \text{topology}$ (F33 Charge-Unit Obstruction, CLOSED-NEGATIVE-under-R1–R3)
can F42 close it?	NO $\rightarrow$ F42 tools are (A, Q) -dimensionless; χ_- is dimensionful (A25's no-go, sharpest form)

Residual (v1.7)	v1.8 disposition
can external math lift it?	NO $\rightarrow$ Dirac/GVW/Freed–Witten fix integers, not the unit e_6 ; external math <i>confirms</i> the obstruction
where does χ_- live?	fully localized $\rightarrow$ F34 master form, residual = membrane charge e_6 , deferred to ZS-F35/F36 (not in corpus)

Final count: the frozen-excess hierarchy generator is **structurally complete**; the single remaining unknown is the dimensionful $\chi_- = e_-^2/(4\pi^2 Z_-)$, a **genuine OPEN the corpus proves current tools cannot close**. New inherited dependencies (beyond v1.7): **ZS-F33 v1.8** (the Charge-Unit Obstruction, F33.8a; $\rho_\Lambda, Z = \frac{1}{2}\chi_- \omega^2$) and **ZS-F34 v1.8** (the six-dimensional charge unit; the residual membrane charge e_6). B3's absolute scale is **not closed** — and F42 registers, as its terminal NON-CLAIM, that no current-corpus tool can close it: the dimensionful e_6 is deferred to the not-yet-existent ZS-F35/F36, exactly where the corpus's charge-unit arc places it. This is the honest end of the frozen-excess line.

References

- [1] M. Henneaux and C. Teitelboim, "The cosmological constant as a canonical variable," Phys. Lett. B **143**, 415 (1984); "The cosmological constant and general covariance," Phys. Lett. B **222**, 195 (1989). [2] R. D. Sorkin, in *Relativity and Gravitation: Classical and Quantum*, Proc. SILARG VII, Cocoyoc, 1990 (World Scientific, Singapore, 1991), pp. 150–173; Int. J. Theor. Phys. **36**, 2759 (1997). [3] M. Ahmed, S. Dodelson, P. B. Greene, and R. D. Sorkin, "Everpresent Λ ," Phys. Rev. D **69**, 103523 (2004), arXiv:astro-ph/0209274. [4] M. Ahmed and R. D. Sorkin, "Everpresent Λ . II. Structural stability," Phys. Rev. D **87**, 063515 (2013), arXiv:1210.2589. [5] N. Zwane, N. Afshordi, and R. D. Sorkin, "Cosmological tests of Everpresent Λ ," Class. Quantum Grav. **35**, 194002 (2018), arXiv:1703.06265. [6] H. Leutwyler and A. V. Smilga, "Spectrum of Dirac operator and role of winding number in QCD," Phys. Rev. D **46**, 5607 (1992). [7] D. Quillen, "Determinants of Cauchy–Riemann operators over a Riemann surface," Funct. Anal. Appl. **19**, 31 (1985). [8] J.-M. Bismut and D. S. Freed, "The analysis of elliptic families. I. Metrics and connections on determinant bundles," Commun. Math. Phys. **106**, 159 (1986); "II. Dirac operators, eta invariants, and the holonomy theorem," Commun. Math. Phys. **107**, 103 (1986). [9] J.-M. Bismut, H. Gillet, and C. Soulé, "Analytic torsion and holomorphic determinant bundles I–III," Commun. Math. Phys. **115**, 49 (1988); **115**, 301 (1988). [10] M. Aizenman, J. L. Lebowitz, and D. Ruelle, "Some rigorous results on the Sherrington–Kirkpatrick spin glass model," Commun. Math. Phys. **112**, 3 (1987). [11] C. Dasgupta and S.-k. Ma, "Low-temperature properties of the random Heisenberg antiferromagnetic chain," Phys. Rev. B **22**, 1305 (1980). [12] D. S. Fisher, "Random antiferromagnetic quantum spin chains," Phys. Rev. B **50**, 3799 (1994). [13] N. Kaloper and A. Padilla, "Sequestering the standard model vacuum energy," Phys. Rev. Lett. **112**, 091304 (2014). [14] N. Kaloper and L. Sorbo, "A natural framework for chaotic inflation," Phys. Rev. Lett. **102**, 121301 (2009). [15] G. Pólya, "Über eine Aufgabe der Wahrscheinlichkeitsrechnung betreffend die Irrfahrt im Straßennetz," Math. Ann. **84**, 149 (1921). [16] A. Khinchin, "Über einen Satz der Wahrscheinlichkeitsrechnung," Fund. Math. **6**, 9 (1924). [17] E. Witten, "Current algebra theorems for the U(1) Goldstone boson," Nucl. Phys. B **156**, 269 (1979).

[18] R. Bousso and J. Polchinski, "Quantization of four-form fluxes and dynamical neutralization of the cosmological constant," JHEP **06**, 006 (2000), arXiv:hep-th/0004134. [19] N. Aghanim *et al.* (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020). [20] E. B. Davies, "Markovian master equations," Commun. Math. Phys. **39**, 91 (1974); "Markovian master equations. II," Math. Ann. **219**, 147 (1976). [21] J. G. Kemeny and J. L. Snell, *Finite Markov Chains* (Van Nostrand, Princeton, 1960), Ch. VI (strong lumpability); cf. G. Lindblad, Commun. Math. Phys. **48**, 119 (1976). [22] K. Kang, *Dimension-Weighted Mediator Semigroups and Observer-Algebra Weights in a Finite-Register de Sitter Model*, ZS-A23 v3.3 (2026), §§4–6 (Theorems A23.14a/14c PROVEN, A23.15 detailed balance, A23.MC size-bias). [23] K. Kang, *Dimension-Weighted Mediator Semigroups and Their Spin-Graded Continuous-Core Lift*, ZS-A24 v2.1 (2026), §14.2 (anti-numerology retraction of the per-edge = $1/\kappa^2$ identification; seam action = π/A). [24] K. Kang, *(Z-Spin spinor / 4π -closure foundations)*, ZS-M3 §6 (the corpus single-seam action π/A); and ZS-F20 v1.1 / ZS-A6 §5.2 (the Z-Telomere winding count $N_{-}(2\pi) = 2\pi/A$, PROVEN 50-digit). [25] K. Kang, *The Cosmological-Constant Absolute Scale in Z-Spin Cosmology — Capstone, Corrected*, ZS-A25 v1.6 (2026), §5 (Escape 2, the Everpresent- Λ $O(1)$ target $\chi_{Z/\alpha_patch} = (3\Omega_{-}\Lambda)^2 = 4.2348$; the four-volume quantum clock v_now , $[\Lambda, \hat{T}_4] = i\hbar$). [26] K. Kang, *The Cosmological Constant as a Quantum State Variable in Z-Spin Cosmology*, ZS-A26 v2.2 (2026) (χ_Z computed on the (3,2,6)/11 bottleneck into the $O(1)$ bracket (0.2, 2.1); α_patch absent, COMPUTED-INCOMPLETE; the matter-sector Brown–Kuchař graph, rank 37, not the Λ -sector operator). [27] M. Henneaux and C. Teitelboim, *The cosmological constant and general covariance*, Phys. Lett. B **222**, 195 (1989) (the unimodular Λ -as-Lagrange-multiplier term underlying A25 Escape 1). [28] W. G. Unruh, *A unimodular theory of canonical quantum gravity*, Phys. Rev. D **40**, 1048 (1989) (the $[H + \Lambda V_4]\Psi = 0$ constraint; the $\hat{T}_4$ /eigenvalue face of A25's $[\Lambda, \hat{T}_4] = i\hbar$). [29] K. Kang, *Z-Spin Boundary Tension as Geometric Dust — a Rank-Absorption No-Go and the ZHCS Program*, ZS-A19 v3.1 (2026), §8.2a (Theorem A19.NG2, the scalar-clock rank-absorption NO-GO and its repair R1 = operator-valued dynamics). [30] K. Kang, *Modular (KMS) Tilt and the Tomita–Takesaki Direction*, ZS-F19 v2.2 (2026) (the modular Hamiltonian $K_{-i} = -\ln \pi_{-i}$; the modular difference $K_{-Y} = -\ln(6/11)$). [31] K. Kang, *The Conditional-Expectation Lift of the Z-Spin Continuous Core — the Loxodromic Seam, and the Compact-Monodromy Reduction*, ZS-F32 v1.5 (2026), F32.9 (CE-local extension uniqueness), F32.24–26 (compact $U(1)$ 3-form Dirac quantization, θ -branch selection $k^*=0$, compact-circle Friedrichs spectral confirmation), F32.27 (the odd topological susceptibility χ_{-}). [32] M. Reed and B. Simon, *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness* (Academic Press, 1975), §X.3 (Friedrichs extension of a semibounded quadratic form). [33] K. Kang, *The Conditional UV Reduction of the Z-Spin Odd Three-Form — the Charge-Unit Obstruction to the Absolute Vacuum Scale*, ZS-F33 v1.8 (2026), F33.8a (the Charge-Unit Obstruction, CLOSED-NEGATIVE-under-R1–R3: flux integrality fixes k but not e^{-2}/Z_{-}); $\rho_{-}\Lambda_Z = \frac{1}{2}\chi_{-}\omega^2$. [34] K. Kang, *The Six-Dimensional Charge Unit of the Z-Spin Three-Form*, ZS-F34 v1.8 (2026) (the master quadratic form for χ_{-} ; the residual membrane charge e_6 deferred to ZS-F35/F36; the rank-2 internal-cycle ansatz $Y_6 = M_4 \times \Sigma_2$). [ZS] Z-Spin Cosmology corpus: ZS-M1; ZS-M8; ZS-M46 v1.4; ZS-F31 v1.4; ZS-F32 v1.5; ZS-F33 v1.8; ZS-F34 v1.8; ZS-F35 v1.5; ZS-F36 v2.1; ZS-F40 v1.2; ZS-F41 v1.1; ZS-A25 v1.6; ZS-A26; ZS-A28; ZS-A30 v2.1; ZS-A32 v1.1 (K. Kang, 2026).

Version History

Full per-revision detail is in §0.1 (one-line summary) and Appendix C (Records 1–11, five-step exploration logs). This history is the audit-standard short form.

v1.0 (March 2026): Initial release under ZS-F41 clause (b): Three-Trap No-Go, Flux-Decomposition and $\sqrt{N_4}$ Excess theorems (PROVEN, MC 1200/1200), (H-FROZEN-GAS), target $c_\chi = 498/(121\omega^2)$. Verification 36/36.

v1.1–v1.7 (March 2026): The structural-closure arc (details Appendix C, Records 4–10; ledgers Appendix C.1). v1.1 closes T-GKLS/ v_q /barrier (F42.12–15); v1.2 inherits the ZS-A23 semigroup (F42.16–18); v1.3 corrects the seam action to π/A (F42.18', O-9 retracted); v1.4 proves c_χ = the A25 Everpresent target (F42.19–20); v1.5 unifies the two residuals as A25's $[\Lambda, \hat{T}_4] = i\hbar$ conjugate pair (F42.21–23); v1.6 closes the finite register with $\Lambda_1 = -\ln(6/11) = K_Y$ (F42.24–27); v1.7 closes the continuous core structurally via F32.9 + F32.26 Friedrichs (F42.28–31). Verification advances 51/51 $\rightarrow$ 93/93. All closures are of the dimensionless operator structure, never the absolute scale.

v1.8 (March 2026): The honest terminus. The dimensionful $\chi(A, Q, \bar{M}_P)$ is a genuine OPEN the corpus proves current tools cannot close: Charge-Unit Obstruction (F42.32, PROVEN inherited), F42-unreachability (F42.33), external math confirms not lifts (F42.34), terminal NON-CLAIM with residual = membrane charge e_6 deferred to ZS-F35/F36 (F42.35). Firewall O-16; gates F-F42.36–38; NC-F42.29–32. Verification 101/101 PASS + 23/23 guards.

v1.9 (March 2026) = v1.8.1 editorial consolidation: No result, theorem, or verification changed. Abstract compressed to the v1.8 terminus (≈ 700 words); per-revision history compressed to one line each (§0.1); §11A–§11H replaced by the unified structural-closure table (§11A) plus the v1.8 terminus (§11B); all superseded per-version ledgers (§12'–§12''''*) moved to Appendix C.1 (Versioned Audit Trail); the body now carries the single §12 Final Residual Ledger. Title adds "Honest Terminus"; "structural closure $\neq$ absolute-scale closure" stated at every load-bearing point. Verification 101/101 PASS + 23/23 guards ([zs_f42_verify_v1_9.py](#), identical checks to v1.8). Zero fitted parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED. (Consolidated from internal Z-Spin Collaboration deep-exploration notes up to v1.8.1.)